



CHARUSAT[®]
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ACADEMIC REGULATIONS & SYLLABUS



First Year B. Tech. (Bachelor of Technology) Programme
(EC) 2025-26

Faculty of Technology & Engineering

Chandubhai S. Patel Institute of Technology

Charotar University of Science and Technology (CHARUSAT)

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CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering



Academic Regulations

Bachelor of Technology (EC) Programme



FACULTY OF TECHNOLOGY AND ENGINEERING ACADEMIC REGULATIONS

Bachelor of Technology Programmes

Academic regulations recommendations are provided to ensure uniform system of education, programmes duration, eligibility criteria for admission, course credits distribution, teaching and examination pedagogy, detailed syllabus with reference material.

1) System of Education

The Charotar University of Science and Technology (CHARUSAT) shall follow Choice based Credit System (CBCS) with Semester pattern at Undergraduate and Master levels. Each semester will be at least of 90 working days. Apart from the programme core courses, provision for choosing University level electives and Programme electives are available under the CBCS.

2) Duration of Programme

i)	Undergraduate programme	(B.Tech.)
	Minimum	8 semesters (4 academic years)
	Maximum	14 semesters (7 academic years)

3) Eligibility for admissions

As enacted by Government of Gujarat/AICTE/UGC from time to time.

4) Mode of admissions

As enacted by Government of Gujarat from time to time.

5) Programme Structure and Credits

As per annexure – 1 attached

6) Attendance

- 6.1. Students are expected to maintain 100% attendance in all courses. However, students may involuntarily have to miss classes due to illness or some family emergency; students are permitted to maintain a minimum attendance of 75% with producing

proof or reason for the absence. In case of medical exigencies, the student/parent should inform the principal immediately through call or by email. Within a week, starting from the day of absence, the proof of medical exigency must be submitted to the Principal's office.

- 6.2. Unauthorized absence will be considered as part of the discretionary 25% for fulfilling the minimum 75% attendance requirement for appearing in the examination.
- 6.3. Students nominated/sponsored by the University to represent in various forums like seminars/conferences/workshops/competitions or taking part in co-curricular/extra-curricular events will be given attendance credit provided the student applies in writing for such a leave in advance and obtains sanction from the Principal of his/her Institute for academic related requests.

7) Course Evaluation

7.1. The performance of every student in each course will be evaluated as follows:

- 7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment. The respective department /institute will conduct the continuous assessment. The course faculty members shall share the pedagogy related to the continuous evaluation with the students.
- 7.1.2 Final end-semester examination shall be conducted by the University through written paper, practical test, oral test, presentation by the student or a combination of these.
- 7.1.3 The weightages of continuous assessment and end-semester university examination in overall assessment shall depend on individual course as approved by Academic Council through Faculty Board and Board of Studies.
- 7.1.4 The performance of candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.
- 7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2. Performance in continuous assessment and end-semester University Examination

- 7.2.1 Minimum performance with respect to continuous assessment as well as end-semester university examination will be an important consideration for passing a course.

- 7.2.2 If a candidate fails to obtain minimum required overall percentage of marks (36%), student has to repeat the examination till the minimum required overall percentage obtained.

8) Grade Point System

1. The total of the internal evaluation marks and end semester examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:
2. Proposed Grading Scheme to awarding letter grade and grade point as per NEP 2020

Letter Grade	Grade Point	Grading Scheme for Mark (In %)
O (Outstanding)	10	96.0-100
A+ (Excellent)	9	86.0-95.9
A (Very Good)	8	76.0-85.9
B+ (Good)	7	66.0-75.9
B (Above Average)	6	56.0- 65.9
C (Average)	5	46.0 – 55.9
P (Pass)	4	36.0 – 45.9
F (Fail)	0	Below 36.0
Ab (Absent)	0	Absent

➤ The minimum passing marks for each pattern of evaluation are 36%

3. The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i

and $i = 1$ to n , n = number of courses in the semester

(ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

9) Award of Class

- ❖ The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
First Class with Distinction	$CGPA \geq 7.0 \ \& \ \leq 10.0$
First class	$CGPA \geq 6.0 \ \& \ < 7.0$
Second Class	$CGPA \geq 5.0 \ \& \ < 6.0$
Pass Class	$CGPA < 5.0$

10) Detention Criteria

- ❖ A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in.

Link: <https://charusat.ac.in/> => Student's Corner => Detention Rules



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering



CHOICE BASED CREDIT SYSTEM

Bachelor of Technology (EC) Programme

Choice Based Credit System

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist Foundation courses, Core courses, Elective courses, Non-credit (audit) courses and SWAYAM MOOCs.

1.1 Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2 Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. Programme Core courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3 Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

1.4 Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will be reflected in Student's Grade Sheet but the grade of the course will not be considered to calculate SGPA and CGPA. Attendance and Course Assessment is compulsory for Non Credit Courses.

1.5 Credit Transfer through SWAYAM MOOCs

CHARUSAT provides credit transfer as per UGC guidelines to all the students from SWAYAM against elective courses. The credit transfer is offered in two modes: (a) Partial credit transfer (b) Full credit transfer.

1.6 Medium of Instruction

The Medium of Instruction will be English.

In consonance with the National Education Policy (NEP) 2020 and the guidelines of the University Grants Commission (UGC), Charotar University of Science and Technology (CHARUSAT) implements the Multiple Exit scheme in their Bachelor of Technology programme.

Facilitating multiple exit options with UG certificate/UG diploma/or UG degree depending upon the number of credits secured. Skill component with progressive enhancement in skills in respective disciplines is introduced in the curriculum right from the 1st year of the program to ensure the student employability after every exit.

Students may be permitted to take a break from the study during the period of study but the total duration for completing the programme shall not exceed 7 years. If student wishes, he/she shall be allowed to re-enter the degree programme within three years and complete the degree programme within the stipulated maximum period of seven years. Table 1 shows the exit qualifications along with credit requirements.

Table 1 Exit Qualifications along with Credit Requirements

ACADEMIC LEVEL	EXIT QUALIFICATION AND CREDITS REQUIRED	NATIONAL CREDIT LEVEL (NCrF)
1st year of UG Degree	UG Certificate will be awarded Minimum 40 credits followed by an exit 4-credit skills enhancement course	4.5
2nd year of UG Degree	UG Diploma will be awarded Minimum of 80 credits followed by an exit 4-credit skills enhancement course	5.0
3rd year of UG Degree	B.Sc. will be awarded Minimum of 120 credits	5.5
4th year of UG Degree	B. Tech. will be awarded Minimum of 160 credits, with minimum of 40 credits each at level	6.0

Guidelines for multiple exit along with awarding UG Certificate, UG Diploma, and Degrees:

A. Exit after First Year

Award: UG Certificate in XXXX

Student must undertake a 4-Credits skill enhancement course before commencement of 2nd year. This shall require 120 hours engagement in the relevant industry/organization. Student shall have an option to continue the 2nd year onward study.

The student shall be awarded with “UG Certificate in XXXX”, with redemption of credits from ABC. Total credits redemption shall be 1st year credits + 4 credits earned in summer.

B. Exit after Second Year

Award: UG Diploma in XXXX

Student must undertake a 4-Credits summer internship/ skill enhancement course before commencement of 3rd year. This shall require 120 hours engagement in the relevant industry/organization. Student shall have an option to continue the 3rd year onward study.

The student shall be awarded with “UG Diploma in XXXX”, with redemption of credits from ABC. Total credits redemption shall be 1st and 2nd year credits + 4 credits earned in summer.

C. Exit after Third Year

Award: B.Sc. Degree in XXXX

The student shall be awarded with “B.Sc. Degree in XXXX”, with redemption of 3 years’ credits from ABC.

4-Credit (120 hrs.) Skills Enhancement Courses:

Level/ Branch	EC
1st year of UG Degree	ECUS110: PCB DESIGNING AND TESTING

Value Added Courses:

Inclusion of 2 credits courses on Community Service/ NSS/NCC/ Sports; and provision to earn extra credits based on undertaking Research/ Academic/Cultural/ and other Developmental activities is introduced.

- The component of 'Value-added Courses' could be enriched to include alternatives that could contribute to the holistic development of the students. In light of this, it is proposed to include a 02 credit Course on Community Service/ NSS/NCC/Sports as a compulsory course in the existing curricula offered across all UG programs at CHARUSAT. It is mandatory for all UG students (2024-25 admission batch) to undertake this course and the course shall not account for the overall CGPA.
- Provision to earn extra credits (**Skill Augmentation** course) based on co-curricular & extra-curricular activities is proposed. It was emphasized that the extra credits could help in fuller realization of the Graduate Attributes laid down by the university as well as could serve as an important ingredient facilitating the progression and recognition of fast learners. It is mandatory for all UG students (2024-25 admission batch) to undertake this course during the study and the credits of the course shall not account for the overall CGPA.

Teaching & Examination Scheme of B.Tech. Programme (EC)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
Teaching & Examination Scheme of B.Tech. Programme (EC)
First Year Semester-I

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
FY Sem-I	MSUD101	Engineering Mathematics-I	4	0	4	4	50	50	0	0	100
	ECUC101	Basics of Electronics Engineering	3	2	5	4	50	50	25	25	150
	ECUC102	Basic C Programming	3	4	7	5	50	50	25	25	150
	ITUS101	ICT Workshop	0	2	2	2	0	0	25	25	50
	PSUD101	Engineering Physics-I	0	2	2	2	0	0	25	25	50
	FTUC101	Foundation Course on Mathematics and Physics	0	2	2	0	0	0	0	50	50
	HSUA101	Communicative English	0	2	2	2	0	0	25	25	50
	CUUV102	Physical Education and Sports	90			2	--	--	--	--	--
	CUUV101	Community Engagement and Sustainable Development	90			2	--	--	--	--	--
	FTXXXX	Skill Augmentation	200			2	--	--	100 Activity Points	--	--

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
Teaching & Examination Scheme of B.Tech. Programme (EC)
First Year Semester-2

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
FY Sem-2	MSUDI02	Engineering Mathematics-II	4	0	4	4	50	50	0	0	100
	ECUCI03	Basics of Digital Electronics	2	2	4	3	50	50	25	25	150
	ECUCI04	Applied Programming	3	4	7	5	50	50	25	25	150
	MEUDI01	Elements of Engineering	3	2	5	4	50	50	25	25	150
	CLUVI01	Environmental Sciences	0	2	2	2	0	0	25	25	50
	PSUDI02	Engineering Physics-II	0	2	2	2	0	0	25	25	50
	HSUSI01-117	Course On Liberal Arts (HS Electives)	0	2	2	2	0	0	25	25	50
	CUUVI02	Physical Education and Sports	90			2	--	--	--	--	--
	CUUVI01	Community Engagement and Sustainable Development	90			2	--	--	--	100	100
	FTXXXX	Skill Augmentation	200			2	100 Activity Points	--	--	FTXXXX	Skill Augmentation

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

B.Tech (EC) Programme

SYLLABI (*SEMESTER-1*)

MSUDI01: ENGINEERING MATHEMATICS-I

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	-	-	4	4
Marks	100	-	-	100	

Prerequisites: Set theory, Function, Limit, Continuity, Differentiability for function of single variable and its uses, Sequence and Series

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Higher-order derivatives and applications	16
2	Infinite Series and Complex numbers	14
3	Matrix Algebra- I	10
4	Partial differentiation	08
5	Applications of Partial Differentiation	12

Total hours (Theory): 60

Total hours (Lab): 00

Total hours: 60

Detailed Syllabus:

1	Higher-order derivatives and applications	16 Hrs	27%
Lagrange's Mean Value Theorem, Local Maxima and Minima of function of one variable Successive differentiation: n^{th} derivative of elementary functions: rational, logarithmic, trigonometric, exponential and hyperbolic, Leibnitz rule for the n^{th} order derivatives of product of two functions, Power series expansion of a function: Maclaurin's and Taylor's series expansion, L'Hospital's rule and related applications, Indeterminate forms.			
2	Infinite Series and Complex numbers	14 Hrs	23%
Tests of convergence of series viz., comparison test, ratio test, root test, Leibnitz test, Complex numbers and their geometric representation, Complex numbers in polar and exponential forms, De Moivre's theorem and its applications, Exponential, Logarithmic, Trigonometric and hyperbolic functions			

3	Matrix Algebra- I	10 Hrs	17%
Definition of Matrix, types of matrices and their properties, Determinant and their properties, Rank and nullity of a matrix, Determination of rank, Solution of a system of linear equations by Gauss elimination and Gauss Jordan Methods			
4	Partial differentiation	08 Hrs	13%
Partial derivative and geometrical interpretation, Euler's theorem with corollaries and their applications, Chain rule, Implicit functions, Total differentials			
5	Applications of Partial Differentiations	12 Hrs	20%
Maclaurin's and Taylor's series expansion in two variables, Tangent plane and normal line to a surface, Maxima and Minima of function of two variables, Lagrange's method of undetermined multiplier, Jacobian, Errors and approximations			

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	find successive differentiation, utilize appropriate theory and computational techniques to construct Taylor's series, use L'Hospital's rule to compute limits of the indeterminate forms.
CO2	Check the convergence of infinite series, perform basic mathematical operations on complex numbers in Cartesian and polar forms, find the n^{th} roots of a complex number and solutions of simple polynomial equations
CO3	find the determinant of a square matrix, evaluate rank and nullity of a matrix, solve system of linear equations by using concept of matrices which are useful in various fields of engineering.
CO4	evaluate partial derivatives including higher order derivatives, solve problems using the chain rules, Euler's theorem with corollaries, implicit function and total differentials.
CO5	expand any function of two variables in ascending power of variables, solve problems using the techniques of multivariable calculus in various branches of engineering.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	2	1	-	2	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	1	-	-	-	-	-	-	-	-	-
CO5	3	2	1	1	1	-	-	-	-	-	-	-	-	-

Correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Recommended Study Material:

❖ **Text Books:**

1. Erwin Kreyszig; Advanced Engineering Mathematics, 8th Ed., Jhon Wiley & Sons, India, 1999.
2. H. K. Dass and Rajnish Verma; Higher Engineering Mathematics, S Chand & Co Pvt Ltd. 2012.
3. B. S. Grewal; Higher Engineering Mathematics, Khanna Publ., Delhi, 2012

❖ **Reference Books:**

1. M. D. Weir *et al.*; Thomas' Calculus, 11th Ed., Pearson Education, 2008.
2. James Stewart; Calculus Early Transcendental, 5th Ed., Thomson India, 2007
3. C. R. Wylie and L. C. Barrett; Advanced Engineering Mathematics. 1982., McGraw-Hill Book Company.
4. Michael D. Greenberg; Advanced Engineering Mathematics. Prentice-Hall, 1988.

URL Links:

1. <https://ocw.mit.edu/ans7870/resources/Strang/Edited/Calculus/Calculus.pdf>
2. <http://nptel.ac.in/courses/111107108/>
3. <http://nptel.ac.in/courses/122101003/>
4. <http://nptel.ac.in/courses/111104085/>

ECUC101: BASICS OF ELECTRONICS ENGINEERING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Basic Electrical Terms and Units	04
2	Electrical Circuit Analysis	07
3	Electrostatic	08
4	Electromagnetism	07
5	AC Fundamentals	05
6	Semiconductor Diode	07
7	Transistor and its characteristics	07

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

Detailed Syllabus:

1	Basics Electrical Terms and Units	04 Hrs	10%
Ohm's law, resistor and its coding, properties, temperature co-efficient of resistance, resistance variation with temperature, examples			
2	Electrical Circuit Analysis	07 Hrs	15%
Kirchhoff's current and voltage law, mesh and nodal analysis, Examples, Series and Parallel circuit connections, star-delta transformation			
3	Electrostatic	08 Hrs	18%
Capacitors, charge and voltage, capacitance, electric fields, electric field strength and electric flux density, relative permittivity, dielectric strength, Examples, Capacitors in parallel and series, Calculation of capacitance of parallel plate and multi plate capacitor, examples			
4	Electromagnetism	07 Hrs	15%
Magnetic field, its direction and characteristics, magnetic flux and flux density, magnetomotive force and magnetic field strength, examples, Faraday's law of electromagnetic			

induction, Fleming's left hand and right hand rule, Lenz law, force on a current carrying conductor, examples, Self and mutual inductance			
5	AC Fundamentals	05 Hrs	12%
AC Waveform and definition of its terms, relation between speed and frequency, Average and RMS value and its determination for sinusoidal wave shapes, examples			
6	Semiconductor Diode	07 Hrs	15%
Semiconductor, Intrinsic Semiconductor, Extrinsic Semiconductor, pn-Junction, Properties of pn-Junction, voltampere characteristics of pn-Junction, Limitations in the Operating Conditions of pn-Junction, Diode equivalent circuit, Diode rectifiers, Half wave rectifier, Efficiency of Half wave rectifier, Full wave rectifier, Efficiency of Full Wave rectifier, Examples Filter, Types of Filter Circuits, Zener Diode, Zener diode as Voltage Regulator, Examples			
7	Transistor and its characteristics	07 Hrs	15%
Transistor , Some Facts about the Transistor , Transistor Symbols , Transistor Connections : Common Base, Common Emitter, Common Collector Connection, Examples 7.2 Characteristics of Common Base and Common Emitter Connection, Measurement of Leakage Current, Operating Regions of Transistor, Commonly Used Transistor Connection, Examples			

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understanding and analysis of basic electrical circuits with independent voltage and current sources.
CO2	CO2 Understanding of Capacitor, working of capacitor and its properties.
CO3	Identify the property of magnetic circuits and understand the laws of emf generation.
CO4	Understanding of basic fundamental concepts as well as practical behaviour of basic semiconductor diode and its applications
CO5	Knowledge of transistor, its characteristics and applications.

Course Articulation Matrix:

	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CO1	3	2	-	-	-	-	-	-	-	-	-	-	1	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	1	
CO3	1	3	-	-	-	-	-	-	-	-	-	-	1	
CO4	1	2	-	2	-	-	-	-	-	-	-	-	3	
CO5	1	2	3	-	3	-	-	-	-	-	-	-	3	

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text Books:**

1. Elements of Electrical Engineering and Electronics by U. A. Patel and R. P. Ajwalia
2. A Text Book of Electrical Technology by B. L. Thareja, S. Chand
3. Principles of Electrical Engineering and Electronics by V. K. Mehta, S. Chand
4. Integrated Electronics by Jacob Millman and Christos C Halkias, Tata McGraw Hill Edition

❖ **Reference Books:**

1. Electrical Technology by Hughes, Pearson Education.
2. Electrical Engineering Fundamentals by Vincent Del Toro, Pearson Education.

❖ **Web material:**

- <https://www.electronics-tutorials.ws/>

ECUCI02: BASIC C PROGRAMMING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	4	-	7	5
Marks	100	100	-	200	

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to 'C' Language.	03
2.	Elements of C	07
3.	Operators and Expression	05
4.	Conditional Statements & Branching	06
5.	Arrays	06
6.	Pointers and Dynamic Memory Allocation	07
7.	Strings	05
8.	User-Defined Functions	06

Total hours (Theory): 45

Total hours (Lab): 60

Total hours: 105

Detailed Syllabus:

1	Introduction to 'C' Language	03 Hrs	05%
Design Methods, Program, Software, Instruction, debugging, compilation and execution of C Program, Difference between Header files & library files, Compiler and Interpreter, Procedure Oriented Language, Importance of C, Basic structure of C, Algorithms & Flowchart.			
2	Elements of C	07 Hrs	10%
Character set, C tokens, Keywords & Identifiers, Data types, Constants, Variables, Declaration of Variables, Assigning Values to Variables, Declaring a variable as Constant, Defining Symbolic constants, Expressions, Statements, Input-Output in C			
3	Operators & Expressions	05 Hrs	18%
Classification of operators: Arithmetic, Relational, Logical, Assignment, Increment / Decrement, Conditional, Bitwise, Special Operators, Unary, Binary and Ternary Operators. Type conversion: Implicit & Explicit, Precedence and Associativity, Various library functions			

4	Conditional Statement and Branching	06 Hrs	23%
Compound statement, if-else, if-else nesting, else-if Ladder, Loops, while statement, do-while statement, for statement, Nesting of Loops, Infinite loops, break statement, continue statement, goto statement, switch			
5	Arrays	06 Hrs	10%
Need of Array, Declaration and Initialization of 1-D Array, Declaration and Initialization of 2-D Array, Memory allocation of 1D and 2D Array, 1-D and 2D array basic programs.			
6	Pointers & Dynamic memory allocation	07 Hrs	12%
About memory, Address Operator, Pointer variables, Declaration of pointer variables, Assigning address to pointer variables, Dereferencing pointer variables, pointer arithmetic, pointer comparisons, pointer to pointer, pointers and 1-D arrays, pointers and 2-D arrays, Array of Pointers, void pointers, Dynamic memory Allocation, malloc, calloc, realloc, free, Dynamic arrays			
7	Strings	05 Hrs	09%
Character arrays, Difference of character array with numeric array and importance of NULL character. Declaration, Initialization and various input and output methods of string, formatted output of string, arithmetic operations on characters. Various functions of string.h: strlen, strcat, strcmp, strcpy, strrev, strstr, etc. Two dimensional character array (table of strings).			
8	User-Defined Function	06 Hrs	13%
Need of modularization, advantages, Introduction to user-defined function, Function Prototype, Function Call or Invocation, return statement, function parameters and Arguments, Actual & Formal Arguments, Categories of functions, Nesting of Functions, Recursion. Array as Function arguments, pointers and functions, Storage Classes: Scope, Life of a variable.			

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Demonstrate problem solving skills by developing algorithms and drawing flowcharts to solve simple problems, Understand the process of compiling and executing a C program and recognize various C tokens and datatypes
CO2	Understanding various programming constructs and applying it for the problems given in hand
CO3	Demonstrate the use of various data structures like array
CO4	Evaluate how pointers are effective in handling arrays, functions and data tables and how pointers support Dynamic memory management
CO5	Applying the concepts of top-down modular programming to decompose problem and a program solution into smaller pieces and Analyse how length of the source program can be reduced by using functions
CO6	Develop C Programs using various methods described above to solve real-world problems

Course Articulation Matrix:

	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CO1	3	1	-	-	-	-	-	1	2	2	1	1	3	1
CO2	3	3	2	-	-	-	-	-	1	-	-	2	3	3
CO3	3	2	2	-	-	-	-	-	1	-	-	2	3	2
CO 4	3	3	3	-	-	-	-	-	2	-	-	3	3	3
CO5	3	3	3	-	-	-	-	-	2	-	-	3	3	3
CO6	3	3	3	-	-	-	-	-	2	-	1	3	3	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text Books:**

1. C in depth, 3th Edition by S. K. Srivastava and D. Srivastava, BPB Publication
2. Let us C, 16th Edition, By Yashwant Kanetkar, BPB Publication
3. C: The Complete Reference, 4th edition, by Herbert Schildt, McGraw Hill Education

❖ Reference Books:

1. C PROGRAMMING (Absolute Beginner's Guide), 3rd Edition, By Greg Perry and Dean Miller, Que Publishing
2. The C Programming Language, 2nd Edition, By Brian W. Kernighan and Dennis Ritchie, Pearson
3. Programming in ANSI C, 8th Edition, By E Balagurusamy, McGrawHill
4. Programming in C, 2nd Edition, By Pradeep Dey and Manas Ghos
5. C How to program, 7th edition, By Deitel&Deitel, Prentice Hall

❖ Web material:

- <https://www.tutorialspoint.com/cprogramming/index.htm>

- <https://www.cprogramming.com/tutorial.html>
- <https://www.programiz.com/c-programming>

❖ Software:

- Turbo C in windows
- GCC compiler in Linux
- VS code
- Code Block

ITUS101: ICT WORKSHOP

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	0	2	-	2	2
Marks	0	50	-	50	

Objective of the Course:

- The main objectives for offering the course ICT Workshop are:
- To explain the fundamentals of computers and peripherals.
- To introduce hardware and software computers basics.
- To deliver concepts and methodology of different parts of the computer and their assembling.
- To brief the students regarding various operating systems installation, commands and scripting in OS.
- To introduce the basic concepts of batch file programming and its uses.

Outline of the Course

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Microprocessor & Computer Organization	4
2	Operating System Installation, Configuration and Commands	4
3	Google Workspace (MOOC Course)	4
4	Hands On GitHub	2
5	Playing with Network	4
6	The basic sensors interfacing using Arduino and ESP32	16

Total Hours: 30

Detailed Syllabus

1	Microprocessor & Computer Organization Introduction to computer Hardware, Internal device of CPU: Motherboard, Processor, Memory, Sound Card, Network Card, Assembly of computer	4 Hours	10%
2	Operating System Installation, Configuration and Commands Windows & Linux Installation with Partition, Installation in Virtual Box, Linux Commands, OS versions with history(Self Study)	4 Hours	10%
3	Google Workspace (MOOC Course) Google Workspace, Email Writing, Excel, Word, Powerpoint, Google Site Do 1 Online Course out of 4 https://www.cloudskillsboost.google/course_templates/90 https://www.aeseducation.com/blog/gsuite-certification https://cloud.google.com/training/workspace https://cloud.google.com/certification/workspace	4 Hours	10%
4	Hands On GitHub What and Why Github, Working collaboratively with GitHub, Working locally using git	2 Hours	5%
5	Playing with Network Signal transmission, Hub/Switch, Cable Crimping, Network Commands & configuration of IP, Advance configuration: Row level, IP Addressing	4 Hours	10%
6	The basic sensors interfacing using Arduino and ESP32 Introduction to Arduino/ESP32 & Sample Program (Blink LED, LED with timer, LED control from Switch) Sensor interfacing with Arduino and ESP32. How to connect and work with different sensors, such as Ultrasonic Sensor, Temperature and humidity, soil moisture, water level indicator with Buzzer etc to Ardunio and ESP32 Board.	16 Hours	60%

	Simple Apps (MIT APP Inverter) Bluetooth module interfacing (whole IoT along with App) Wi-Fi on NodeMCU/ESP32 Raspberry pi +Grove Pi (Sound Sensor, Rotary Angle, LCD Display) Raspberry pi face detection Project		
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Student Learning Outcomes

At the end of the course, the students will be able to

CO1	A student will be having the basic knowledge of computer architecture, peripherals and all the hardware and software basics.
CO2	A student will be able to understand hardware requirement for operating system and able to install it on a machine.
CO3	Analyze and generate the different parsing techniques..
CO4	A student will become familiar with command line interface of Windows and Linux.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	-	-	-	2	-
CO2	3	1	-	-	-	1	-	-	-	-	-	-	2	-
CO3	3	2	2	-	2	-	-	-	-	-	-	-	3	1
CO4	3	-	-	-	1	1	-	-	-	-	-	-	2	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Student Learning Outcomes:

By taking this course,

- A student will be having the basic knowledge of computer architecture, peripherals and all the hardware and software basics.
- A student will be able to understand hardware requirements for operating systems and be able to install it on a machine.
- Students will be able to work with basic Network Instruments and be able to configure computers in a simple Network.
- Students will be able to make small projects on IoT and be aware about the various equipment which are useful to create projects in the upcoming semester.
- Provide hands on sessions on GitHub so they can upload their project on an online Repository that helps to showcase it in their Resume/CV.

Recommended Study Material:

Reference Books:

1. The Complete PC Upgrade and Maintenance Guide, 16th Edition, Mark Minasi, Quentin Docter, Faithe Wempen, SYBEX publication
2. IBMPC And Clones Govindarajulu, Tata McGraw Hill

Web Materials:

1. https://www.cloudskillsboost.google/course_templates/90
2. <https://www.aeseducation.com/blog/gsuite-certification>
3. <https://cloud.google.com/training/workspace>
4. <https://cloud.google.com/certification/workspace>
5. [Introduction to Git and GitHub | Coursera](#)

PSUD101: ENGINEERING PHYSICS - I

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	50	-	50	

Outline of the Course:

Unit No.	Title of the Unit	Minimum number of Hours
Unit 1	Mechanics Lab	16
Unit 2	Thermodynamics Lab	14

Total hours (Theory): 00

Total hours (Lab): 30

Total hours: 30

Detailed Syllabus:

Unit	Name	Hours
1.	Mechanics	16 hours (55%)
1.1	Uncertainties in Measurements: Sources and estimation of errors, accuracy and precision, systematic error, random error, Significant figure and round off, error propagation	
1.2	Laws of Motion: Frames of reference. Newton's Laws of motion, Dynamics of a system of particles, Centre of Mass, Projectile motion	
1.3	Collisions: Elastic and inelastic collisions between particles	
1.4	Momentum and Energy: Conservation of momentum, Work and energy, Conservation of energy. Motion of rockets	
1.5	Rotational Motion: Angular velocity and angular momentum. Torque. Conservation of angular momentum	
1.6	Elasticity: Hooke's law - Simple Stress and Strain: Introduction, Normal and Shear stresses, Stress- Strain Diagrams for ductile and brittle material, Elastic Constants	
No. of Experiment 1. Understanding Errors and Uncertainties in the measurements 2. Conservation of Energy, Hook's law 3. Young modulus and elasticity		

4. To determine g by Bar Pendulum 5. To determine g Kater's Pendulum.		
2.	Thermodynamics:	14 hours (45%)
2.1	Zeroth Law of thermodynamics and temperature	
2.2	First law, second, third law and internal energy, conversion of heat into work	
2.3	Various Thermo dynamical Processes, Enthalpy, Gibbs, Helmholtz and Internal Energy functions,	
2.4	Transport Phenomena: Viscosity, Conduction and Diffusion	
2.5	Applications to specific heat of gases and metals	
2.6	Blackbody radiation, Spectral distribution, Derivation of Planck's law	
No. of Experiment 1. Specific Heat of Metals 2. Thermal conductivity of materials by Searl's apparatus 3. Heat Transfer and Newton's Law of Cooling 4. Radiation from a black body: Stefan-Boltzmann Law		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Students will be able to apply and demonstrate the concepts of mechanics to practical engineering problems.
CO2	Understand the basic concepts of thermodynamics such as internal energy, thermodynamic properties, transport phenomena, blackbody radiation.

Course Articulation Matrix:

	PO1	PO 2	PO3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	3	1	2	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	3	1	1	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
PSUD10 1	3.00	3.00	2.00	1.50	1.00	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text Books:

1. Physics for Scientists and Engineers by Randall D. Knight, 4th Edition, Pearson
2. University Physics by Hugh D. Young, Roger A. Freedman and A. Lewis Ford, 13th Edition, Pearson
3. Physics by John D. Cutnell & Kenneth W. Johnson, 8th Edition, John Wiley & Sons, Inc.

❖ Reference Books:

1. An introduction to mechanics, D. Kleppner, R.J. Kolenkow, 1973, McGraw-Hill.
2. University Physics, Ronald Lane Reese, 2003, Thomson Brooks/Cole.
3. Physics, Resnick, Halliday and Walker 8/e. 2008, Wiley.

❖ Web material:

1. Uncertainty and error in measurement - <https://www.youtube.com/watch?v=pWEflsClyTk>
2. Vernier Callipers principle and description (Introduction)- <https://www.youtube.com/watch?v=ySRN3yuZUT0>
3. Hooke's law - <https://www.youtube.com/watch?v=PWQm4ynYVSE>
4. Law Of Conservation Of energy in a Simple Pendulum - https://www.youtube.com/watch?v=5lRCyBr_nGk, <https://www.youtube.com/watch?v=QlViWmQcwpQ>
5. Young's modulus of the material of a beam by method of bending of beam - <https://www.youtube.com/watch?v=iUhfstfI0rk>
6. Bar pendulum - https://www.youtube.com/watch?v=3uZ_Boyt_AI
7. Kater's Pendulum - <https://www.youtube.com/watch?v=TxbDyv17Jfs>
8. Specific Heat of Metals - <https://www.youtube.com/watch?v=8gHFAL2990U>
9. Thermal conductivity of materials by Searl's apparatus - <https://www.youtube.com/watch?v=qKhherqhPfY>
10. Newton's Law of Cooling - <https://www.youtube.com/watch?v=1C9o6ikJlR8>
11. Radiation from a black body: Stefan-Boltzmann Law - <https://www.youtube.com/watch?v=riRsMfNmieM>

FTUC101: FOUNDATION COURSE ON MATHEMATICS AND PHYSICS

Hours:

Teaching Scheme		Total
Hours/week	2	2
Marks	50	50

Examination Scheme:

Quiz: 50 Marks

Objective of the Course:

- The major objective of the course is to strengthen the foundation knowledge of mathematics and physics, which are prerequisites in engineering.
- To clear the concepts of mathematics and physics with interactive modules.

Detailed Syllabus:

Modules of Mathematics

11 th GRADE		
Sr. No.	Topics	Subtopics
1	Complex numbers and quadratic equations	1. Complex numbers 2. De Moivre's theorem and rotation of complex number
2	Sequences and Series	1. Sequences and series 2. Arithmetic progression 3. Geometric progression 4. Sum of some special series
3	Probability	Probability
12 th GRADE		
4	Matrices	1. Types of matrices 2. Operation on Matrices 3. Inverse of matrix elementary operations 4. Special types of matrices 5. Properties of Matrices
5	Continuity and differentiability	1. Differentiability 2. Differentiation of composite function

		3. Mean value theorem
6	Application of derivatives	1. Rate of change of quantities 2. Increasing and decreasing functions 3. Tangents and Normals 4. Maxima and minima
7	Integrals	Integration by parts
8	Applications of integral	1. Area under simple curve 2. Area between two curves
9	Probability	1. Conditional probability 2. Baye's theorem 3. Binomial distribution

Modules of Physics

Sr. No.	Topic	Content
11 th GRADE		
1	Mechanical Properties of Solids	1. Elastic behavior of materials 2. Elastic moduli
2	Mechanical Properties of Fluids	3. Hydrostatics and Bernoulli's principle 4. Viscosity 5. Surface tension
3	Thermal Properties of Matter	6. Heat and temperature 7. Change of state and Newton's law of cooling
4	Oscillators	8. Introduction to Simple Harmonic Motion (Solution of second order differential order) 9. Dynamics of simple harmonic motion 10. Damped forced and resonant oscillators
5	Thermodynamics	11. Zeroth and the first law of thermodynamics 12. Isothermal and adiabatic process 13. Heat engine and the second law of thermodynamics
12 th GRADE		
6	Ray Optics	14. Reflection of light by spherical mirrors 15. Refraction, total internal reflection (TIR), refraction at spherical surfaces, and refraction through a prism. 16. Dispersion by a prism, Rainbow, and scattering of light 17. Eye, microscope, and telescope
7		18. Band model of metal, semiconductors, and insulators

	Semiconductor Electronics Material Devices and Simple Circuits	19. Semiconductor diodes and their application
		20. Transistor action and characteristics
		21. Transistor as an amplifier and oscillator
		22. Transistor as a switch and logic gates
8	Communication Systems	23. Basics elements of communication systems
		24. Propagation of electromagnetic waves
		25. Modulation

Total hours: 30

Instructional Methods and Pedagogy

- The contents of the course will be installed on the student's device (Mobile/Laptop).
- Students have to explore the contents by his/her self on SARAS-3D.
- Students can check the self-paced learning outcome by attending the quiz available after the topic/chapter.
- Course mentors will be assigned to clear any difficulty in understanding the topic
- One external exam (Quiz based) will be conducted. The students have to secure a minimum 45% of marks in the quiz to clear the course.

Student Learning Outcomes:

At the end of the course, the students will acquire fundamental knowledge regarding Mathematics and Physics. This will be a stepping stone for the technical uplift of knowledge and skill of the student.

Recommended Study Material:

Textbooks:

1. NCERT Book of Mathematics for 11th and 12th standard.
2. NCERT Book of Physics for 11th and 12th standard.

HSUA101: COMMUNICATIVE ENGLISH

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	02	2
Marks	--	50	--	50	

Pre-requisite:

- Basic knowledge of English Language & Communication

Objectives of the Course:

- To make students aware of and give them first-hand experience of Communicative English
- To make the students aware of the different communicative functions of English and help them communicate using those functions
- To assist and guide the learners to be able to listen to and to read a text and identify specific and global information
- To assist and guide the learners to be able to speak and to write clearly and effectively

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Communicative English	03
2.	Communication Functions and Grammar	06
3.	Communication Skills I – Listening	03
4.	Communication Skills II – Reading	03
5.	Communication Skills III – Speaking	06
6.	Communication Skills IV – Writing Composition	06
7.	Developing Vocabulary	03

Total hours (Practical): 30

Total hours (Lab) : --

Total hours : 30

Detailed Syllabus:

1.	Introduction to Communicative English	03 Hours	10%
	English as a Window Language; Varieties of English: British English, American English, Indian English; Language Variations;		

	Importance of English for Academic and Professional Development; Strategies for Language Acquisition; Formal VS Informal English		
2.	Communication Functions and Grammar	6 Hours	20%
	Communicative Grammar: Usage of parts of speech in communicative way, Sentence Structure & Tense, Active Passive & Direct-Indirect Greeting and Introducing; Making Requests and Asking for Information; Expressing Likes and Dislikes; Seeking Permission; Giving and Taking Advice; Describing People, Place, Things; Retelling Past Events; Comparing and Contrasting; Persuading; Describing Cause and Effect		
3.	Communication Skills I – Listening	3 Hours	10%
	Importance of Listening as a Language Skill; Basic Listening Skills; Types of Listening; Barriers to Listening; Strategies for Effective Listening; Listening Practice		
4.	Communication Skills II – Reading	3 Hours	10%
	Importance of Reading as a Language Skill, Reading Strategies: Skimming, Scanning, Intensive Reading, Extensive Reading, Strategies for Effective Reading Comprehension, Reading Practice		
5.	Communication Skills III – Speaking	6 Hours	20%
	Importance of Speaking as a Language Skill, Basic Speaking Skills; Paralanguage for Effective Speaking; Strategies for Oral Communication; Extempore and Public Speaking & Presentation		
6.	Communication Skills IV – Writing Composition	6 Hours	20%
	Importance of Writing as a Language Skill; Process of writing: Prewriting, Drafting, Revision, Editing, Publication; Seven C's of Writing; Sentence Construction – Complex, Compound, Paragraph Development; Letter Writing (Academic Context)		
7.	Developing Vocabulary	3 Hours	10%
	High Frequency Vocabulary (Everyday and Academic use); Words Often Confused and Misused; Useful Phrasal Verbs, Idioms and Proverbs, Homonyms and Homographs; Lexical Range, Word Games		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Communicate with people using English language functions including greetings, introductions, making and responding to requests, suggestions, invitations and apologies, conducting simple transactions in shops and offices, asking for and giving directions, etc
CO2	Go through a text and identify specific and global information.
CO3	Become more knowledgeable about speaking strategies and speak effectively using appropriate words, expression, tone and pronunciation.

CO4	Be aware about various reading strategies and read and comprehend academic and non-academic prose (text).
CO5	Write systematically using nuances of writing
CO6	Express their opinion and likes and dislikes, advice and convince others in a more polite and accepted way.

Evaluation Scheme:

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Recommended Study Material:

❖ Text book:

1. Sanjay Kumar and PushpLata (Second Edition, 2015), Communication Skills, Oxford University Press, New Delhi
2. M V Rodriques (2013), Effective Business Communication, Concept Publishing Company (P) Ltd., New Delhi
3. Krishna Mohan and Meera Banerji (2010), Developing Communication Skills, Macmillan Publications India Ltd., New Delhi

❖ Reference book:

1. Mohan and Meenakshi Raman (2006), Effective English Communication, Mcgraw-Hill Publishing Company Limited, New Delhi
2. Geoffrey Leech & Jan Swartvik (1994), A Communicative Grammar of English, Longman Publications, New York
3. Jones Leo (1979), Functions of English, Cambridge University Press, UK
4. European Journal of Language and Literature Studies Vol.1 Nr, 1 April 2015
5. English for Academic Purpose: A Tool for Enhancing Students' Proficiency in English Language Skills

❖ Web material:

1. <https://www.futurelearn.com/courses/language-assessment>
2. <https://www.coursera.org/learn/importance-of-listening?#syllabus>
3. <https://www.futurelearn.com/courses/english-academic-study>
4. <https://www.coursera.org/learn/grammar-punctuation>
5. <https://www.coursra.org/specializations/improve-english>

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

B.Tech (EC) Programme

SYLLABI (*SEMESTER-2*)

MSUDI02: ENGINEERING MATHEMATICS-II

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	-	-	4	4
Marks	100	-	-	100	

Prerequisites: Basic calculus, mean, median, mode, standard deviation, combinatorial probability

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	First order and First degree Ordinary Differential Equations	10
2	Higher Order Ordinary Linear Differential Equations	12
3	Partial Differential Equations and Applications	08
4	Matrix Algebra –II	10
5	Improper and Multiple Integrals	12
6	Probability and Statistics	08

Total hours (Theory): 60

Total hours (Lab): 00

Total hours: 60

Detailed Syllabus:

1	First order and First degree Ordinary Differential Equations	10 Hrs	17%
Formation of Ordinary Differential Equation, Concept of general and particular solution, Initial value problems, Solutions of first order and first degree differential equations: Linear, Bernoulli, Exact and non-exact differential equations			
2	Higher Order Ordinary Linear Differential Equations	12 Hrs	20%
General Solution of Higher Order Ordinary Linear Differential Equations with Constant coefficients, Methods for finding particular integrals viz. variation of parameters and undetermined coefficients, Linear Differential Equation of higher order with variable coefficients: Legendre's Equations (Special case: Cauchy-Euler equation), System of simultaneous first order linear differential equations			

3	Partial Differential Equations	08 Hrs	13%
Formation of Partial Differential Equation, Lagrange's Linear Differential Equations, Special types of Nonlinear First Order Partial Differential Equation			
4	Matrix Algebra –II	10 Hrs	17%
Revision of matrices and determinant, Eigenvalues and Eigenvectors of matrices, Eigenvalues and Eigenvectors of special matrices, Cayley-Hamilton's Theorem and its applications, Crout's method of LU decomposition			
5	Improper and Multiple Integrals	12 Hrs	20%
Introduction to Improper integrals, Definitions and properties of Gamma, Beta and Error functions, Evaluation of double integrals, Change of order of double integration, Transformation to polar coordinates, Applications of double integrals: Area, Evaluation of triple integrals, Transformation cylindrical coordinates, Applications of triple integrals: Volume			
6	Probability and Statistics	08 Hrs	13%
Elementary Probability: Joint and Conditional probability and Bayes theorem, Discrete Probability Distributions: Binomial and Poisson, Continuous Probability Distributions: Exponential and Normal			

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	formulate models of natural phenomena using differential equations and find its solution using standard methods
CO2	identify, analyze and subsequently solve physical problems analytically whose behaviour can be described by linear and nonlinear differential equations.
CO3	find and explain significant of Eigenvalues and Eigenvectors of a square matrix, use Cayley-Hamilton's theorem to find inverse and power of a square matrix, construct LU decomposition of a square matrix.
CO4	use advanced techniques to evaluate improper integrals, apply multiple integrals to find area and volume in engineering field.
CO5	understand and solve the problems using probability axioms, rules and Bayes theorem, use distributions such as Binomial, Poisson, Exponential and Normal to solve real world problems.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	1	-	-	-	-	-	-	-	-	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO5	2	2	1	2	1	1	1	-	-	-	-	-	-	-

Correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Recommended Study Material:

❖ Text Books:

1. Erwin Kreyszig; Advanced Engineering Mathematics, 8th Ed., Jhon Wiley & Sons, India, 1999.
2. H. K. Dass and Rajnish Verma; Higher Engineering Mathematics, S Chand & Co Pvt. Ltd.
3. Sheldon Ross; A first course in probability. Pearson, 2014.
4. B. S. Grewal; Higher Engineering Mathematics, Khanna Publ., Delhi, 2012

❖ Reference Books:

1. M. D. Weir et al; Thomas' Calculus, 11th Ed., Pearson Education, 2008.
2. James Stewart; Calculus Early Transcendental, 5th Ed., Thomson India, 2007
3. C. R. Wylie and L. C. Barrett; Advanced Engineering Mathematics. 1982, McGraw-Hill Book Company.
4. Michael D. Greenberg; Advanced engineering mathematics. Prentice-Hall, 1988.
5. R. V. Hogg, E. A. Tanis and D. L. Zimmerman; Probability and Statistical Inference, 9th edition, Prentice Hall, 2015.
6. Zafar Ahsan; Differential Equations and Their Applications, ϕ Learning, Pvt Ltd, Third Edition (2017).

❖ URL Links:

1. <http://nptel.ac.in/courses/122107037/>
2. <http://nptel.ac.in/courses/111107108/>
3. <http://nptel.ac.in/courses/122103012/>
4. <http://nptel.ac.in/courses/122104018/>
5. <http://nptel.ac.in/courses/111106100/>
6. <http://nptel.ac.in/courses/122101003/>
7. <https://ocw.mit.edu/courses/mathematics/18-02-multivariable-calculus-fall-2007/lecture-notes/>
8. <https://nptel.ac.in/courses/111105041/>

ECUCI03: BASICS OF DIGITAL ELECTRONICS

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2	2	4	3
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Digital Number System	08
2.	Logic Gates & Boolean Algebra	10
3.	Combinational Circuits	08
4.	Introduction to Sequential Circuits	04

Total Hours (Theory): 30

Total Hours (Lab): 30

Total Hours: 60

Detailed Syllabus:

1.	Digital Number System	08 Hours	26%
Fundamentals of Digital & Analog Systems; Base-r Numbers: Decimal, Binary, Octal, and Hexadecimal Numbers; Number-base Conversions; Digital Arithmetic: Binary addition and subtraction, base-r addition and subtraction; Representation of Negative Numbers; Complements of Numbers: 1's and 2's complement, (r-1)'s and r's complement; Binary Codes: Binary Coded Decimal (BCD), Excess-3 Code, Gray Code and other codes (Alphanumeric, error detection, error correction); BCD addition.			
2.	Logic Gates & Boolean Algebra	10 Hours	34%
Digital Logic Gates: Basic Logic Gates (AND, OR, NOT), Universal Logic Gates (NAND, NOR), Exclusive Logic Gates (XOR, XNOR); Integrated Circuits: Types of ICs, Digital ICs, 7400-series ICs; Boolean Algebra: Basic and Axiomatic definition, Basic Theorems and Properties; Boolean Functions: Reduction using Algebraic Manipulation; Canonical and Standard Forms: Minterms and Maxterms, Sum of Minterms and Product of Maxterms, SOP and POS forms; Minimization of Boolean Expression using K-Map (Up to 4 Variable			

K-Map), Don't-Care Conditions; NAND and NOR implementation, Other Two Level implementation (AOI and OAI Logic).			
3.	Combinational Circuits	08 Hours	26%
Binary Adder-Subtractor: Half Adder and Full Adder, Half Subtractor and Full Subtractor; Decoder, Encoder, Priority Encoder; Multiplexers & Demultiplexer; Code Converters: BCD to/from XS3 Code, Binary to/from Gray Code.			
4	Introduction to Sequential Circuits	04 Hours	14%
Introduction of Sequential Circuit; Storage Elements: Latches, Flip-Flops; Triggering of Flip-flops; Flip Flop with Truth Table and Characteristic Table: SR, D, JK and T; Flip Flop Excitation Table; Conversion of one Flip Flop to other.			

Course Outcomes (CO):

At the end of the course student will be able to:

CO1	Learn the basics of number systems and conversions.
CO2	Understand the basics of logic gates, Boolean algebra, and K-map to deduce the Boolean expressions.
CO3	Design and implement different types of combinational logic circuits.
CO4	Analyze the working of sequential logic circuits.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3	-	-	-	1	1	-	1
CO2	3	3	3	3	3	2	-	-	3	3	3	2
CO3	3	3	3	3	3	-	2	-	3	2	1	2
CO4	3	3	3	3	3	-	2	-	3	2	1	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Text Books:

1. Digital Design, 4th Edition, M. Morris Mano and Michael D. Ciletti, PHI Publication
2. Digital Principles and Application, Anand Kumar, 4th Edition, PHI Publication

❖ Reference Books:

1. Digital Logic and Computer Design, M Morris Mano, PHI Publication

2. Digital System Design Using VHDL, Charles H. Roth, Cengage Publication

❖ Web Materials/ Reading Material:

1. <https://pages.uoregon.edu/rayfrey/DigitalNotes.pdf>
2. <https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/>
3. Hand Outs, Assignments, Question Banks

❖ Software Simulators:

1. LTSpice
2. <https://www.tinkercad.com/>

ECUC104: APPLIED PROGRAMMING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	4	-	7	5
Marks	100	100	-	200	

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Structure and Union	10
2	File Management in 'C'	05
3	C Pre-processor	04
4	Introduction to Object Oriented Programming	05
5	Classes and objects	07
6	Constructor and Destructors	04
7	Operator Overloading and Polymorphism	05
8	Inheritance and Encapsulation	05
	Total hours (Theory):	45
	Total hours (Lab):	60
	Total hours:	105

Detailed Syllabus:

1	Structure and Union	10 Hrs	05%
Defining a Structure, Declaring Structure Variables, Initialization Of Structure Variables, Accessing Members of a Structure, Assignment of Structure Variables, Storage of Structures in Memory, Size of Structure, Array of Structures, Arrays Within Structures, Nested Structures, Pointers to Structures, Pointers Within Structures, Structures And Functions, Self-Referential Structures, Definition of Union, Difference Between Structure and Union			
2	File Management in 'C'	05 Hrs	10%
Text And Binary Modes of Files, Concept Of Buffer, Structure of a General File Program, Creating or Opening files, Errors in Opening Files, Closing Files, Writing to Files and Reading from Files			

3	'C' Pre-processor	04 Hrs	18%
Importance of #define, Macros with Arguments, Nesting in Macros, Problems with Macros, Macros Vs Functions			
4	Introduction to Object Oriented Programming	05 Hrs	23%
Basic concept of object-oriented Programming, Benefits of Object Oriented Programming, Difference between object oriented language and procedure oriented language, Introduction to C++, Applications of C++, Demonstration of simple C++ Program			
5	Classes and objects	07 Hrs	10%
Limitation of C structure, declaring class and defining member function, Making outside function inline, Nesting member function, Private member function, Creating Objects, Accessing Class Members, Memory allocation of objects, Arrays of Objects, Object as a function argument			
6	Constructor and Destructors	04 Hrs	12%
Introduction to Constructors, Parameterized Constructors, Multiple Constructors in class, Destructors			
7	Operator Overloading and Polymorphism	05 Hrs	09%
Introduction, Defining Operator overloading, overloading unary and binary operators, Function Overloading, Concept of Polymorphism			
8	Inheritance and Encapsulation	05 Hrs	13%
Introduction to Inheritance, Defining a derived class, Example of Single, Inheritance, Public and private inheritance. Multilevel, multiple Inheritance, Introduction to Encapsulation, Access Private Members of a Class			

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Demonstrate the use of Structure and Union
CO2	Understand the concept of File Operation and C Pre-processor
CO3	Comprehend the difference between Procedure Oriented Programming and Object Oriented Programming
CO4	Recognize and apply features of object oriented programming such as encapsulation, polymorphism, inheritance and operator overloading based on object identity of respective class
CO5	Develop C and C++ Programs using various methods described above to solve real-world problems

Course Articulation Matrix:

	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CO1	3	1	1	1	-	-	-	-	-	1	-	1	3	1
CO2	2	3	3	2	3	-	-	-	2	-	-	-	2	3

CO3	3	3	3	2	-	-	-	-	-	-	-	-	3	3
CO 4	2	3	3	3	3	1	1	-	2	-	-	-	2	3
CO5	3	3	2	3	2	1	1	-	2	-	1	2	3	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text Books:

1. C in depth, 3th Edition by S. K. Srivastava and D. Srivastava, BPB Publication
2. Object Oriented Programming with C++, E. Balagurusamy, (Tata –MGHill)

❖ Reference Books:

1. Test your C++ Skills, Y. Kanetkar, BPB Publications

❖ Web material:

- https://www.w3schools.com/cpp/cpp_classes.asp
- www.cprogramming.com/tutorial.html
- www.tutorials4u.com/c/
- <http://www.stroustrup.com/C++.html>

❖ Software:

- Turbo C in windows
- GCC compiler in Linux
- VS code
- Code Block

MEUD101: ELEMENTS OF ENGINEERING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	0	5	4
Marks	100	50	0	150	

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
Part: A		
1	Fundamentals of Engineering Drawing	05
2	Projections & Section of Solid	05
3	Orthographic Projection	05
4	Isometric Projections	05
5	Computer Graphics	03
Part: B		
6	Introduction of Mechanical Engineering	02
7	Steam and Steam Generator	03
8	Internal Combustion Engines	03
9	Refrigeration and Air Conditioning Systems	03
Part: C		
10	Scope of Civil Engineering	02
11	Introduction to Surveying	04
12	Elements of building Construction	05

Total hours (Theory): 45

Practical Hours: 30

Total hours 75

Detailed Syllabus:

Part: A			
1	Fundamentals of Engineering Drawing	05 Hours	11 %
1.1	Importance of engineering drawing, drawing instruments and materials, BIS and ISO		
1.2	Different types of lines used in engineering practice, methods of projections as per SP 46-1988.		
1.3	Engineering scale.		
1.4	Engineering curve.		
2	Projections & Section of Solid	05 Hours	11 %
2.1	Projection of solids		
2.2	Sectional view		
2.3	True shape of Sections		
2.4	Auxiliary Inclined Plane (AIP), Auxiliary Vertical Plane (AVP)		

3	Orthographic Projection	05 Hours	11 %
3.1	Principle projection		
3.2	Methods of first and third angle projection with examples / problems		
4	Isometric Projections	05 Hours	11 %
4.1	Terminology, Isometric scale		
4.2	Isometric view and Isometric projection with examples / problems		
5	Computer Graphics	03 Hours	6 %
5.1	Introduction of computer graphics		
5.2	Demonstration of CAD Modeling software		
5.3	Training of Fusion 360 software		
Part: B			
6	Introduction of Mechanical Engineering	02 Hours	4 %
6.1	Prime movers and its types, Sources of energy		
6.2	Basic terminology: Force and mass, Pressure, Work, Power, Energy, Heat, Temperature, Units of heat, Specific heat capacity, Interchange of heat, Change of state, Internal energy, Enthalpy, Entropy, Efficiency		
6.3	Zeroth Law and First Law of Thermodynamic, Boyle's law, Charle's law and Combined gas law, Relation between Cp and Cv		
7	Steam and Steam Generator	03 Hours	7 %
7.1	Introduction to steam formation and its types		
7.2	Introduction to steam table		
7.3	Boiler definition and its classification		
7.4	Cochran boiler.		
8	Internal Combustion Engines	03 Hours	7 %
8.1	Introduction		
8.2	Basic terminology of I.C. engine		
8.3	Types of I. C. engines		
9.	Refrigeration and Air Conditioning Systems	03 Hours	7 %
9.1	Introduction to refrigeration and air conditioning		
9.2	Basic terminology, Principal and application of refrigeration		
9.3	Vapour compression refrigeration system,		
9.4	Window and split air conditioning systems		
Part: C			
10.	Scope of Civil Engineering	02 Hours	5 %
10.1	Scope of Civil Engineering,		
10.2	Branches of civil engineering,		
10.3	Role of civil engineer		
11.	Introduction to Surveying	04 Hours	9 %
11.1	Definition of surveying,		
11.2	Objects of surveying, Uses of surveying,		
11.3	Primary divisions of surveying, Principles of surveying,		
11.4	List of classification of surveying, Definition: Plan and Map,		
11.5	Scales : Plain scale and Diagonal scale, Conventional Symbols		
	Introduction to linear and angular measurements, Concepts of land profiling		
12.	Elements of building Construction	05 Hours	11 %
12.1	Types of building, Design loads,		
12.2	Building components (super structure and substructure),		

12.3	Principles of Planning,		
12.4	Basics Requirements of a building Planning,		
12.5	Types of Residential Building,		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Describe the fundamentals of engineering drawing, engineering scale and engineering curve.
CO2	Use the concept of projection and section of solids for given applications.
CO3	Visualize three-dimensional of engineering components through orthographic, sectional orthographic and isometric drawing and use the computer for geometric modeling.
CO4	Describe the fundamental principles of mechanical engineering and different mechanical system.
CO5	Explain the importance of civil engineering and land surveying.
CO6	Describe the different building components, building planning and design of residential building.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	3	-	-	-	-	-	-	-	-	-	-
CO4	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	2	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	1	2	-	-	-	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. N. D. Bhatt & V. M. Panchal, “Engineering Drawing”, Charotar Publishing House Pvt. Ltd.
2. P. J. Shah, “Engineering Graphics”, S. Chand Publishing & Co.
3. P.S.Desai, S.B.Soni, “Elements of Mechanical Engineering”, Atul Prakashan, Ahmedabad
4. S.M.Bhatt, H.G.Katariya, J.P.Hadiya, “Elements of Mechanical Engineering”, Books India Publication, Ahmedabad.
5. Khasia R.B. and Shukla R.N., “Elements of Civil Engineering”, Mahajan Publication.
6. Punamia B.C., “Surveying”, Vol. I & II.

❖ Reference book:

1. P.B. Patel & P.D. Patel, "Engineering Graphics", Mahajan Publishing House.
2. Arunoday Kumar, "Engineering Graphics", Tech-Max Publication.
3. M.L. Agrawal & R.K. Garg, "Engineering Drawing", Vol. I, Dhanpatrai & Co.
4. Dr. Sadhu Singh, "Elements of Mechanical Engineering", S.CHAND Publication, New Delhi
5. V.K.Manglik, "Elements of Mechanical Engineering", PHI Learning, Delhi.
6. Kandya Anurag, "Elements of Civil Engineering", Charotar Publishing House Pvt. Ltd.
7. Kanetkar T.P. & Kulkarni S.V., "Surveying and Levelling", Vol. I & II.

❖ **Web Materials:**

1. <http://nptel.ac.in/courses/112103019/>
2. <http://nptel.ac.in/downloads/112105125/>
3. <http://nptel.ac.in/syllabus/syllabus.php?subjectId=105104101>
4. <http://nptel.ac.in/courses/105107122/>
5. <https://law.resource.org/pub/in/bis/S01/is.sp.46.2003.pdf>

CLUV101: ENVIRONMENTAL SCIENCES

Credit And Scheme:

Teaching Scheme	Theory/ Practical	Total	Credit	Evaluation Scheme				
				Theory		Practical		Total
Hours/week	2	2	2	Internal	External	Internal	External	
Marks	50	50		--	--	25	25	50

Outline Of The Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Multidisciplinary nature of environmental Sciences	03
2	Environmental Pollution	07
3	Ecology & Ecosystems	05
4	Biodiversity and Conservation	04
5	Natural Resources	04
6	Human Communities and the Environment	07

Total Hours (Theory): 00

Total Hours (Lab): 30

Total Hours: 30

Detailed Syllabus:

1.	Multidisciplinary nature of environmental studies	03 Hours	11%
1.1	Definition, scope and importance		
1.2	Earths-Evolution and Systems		
1.3	Components of the Environment: Biotic, Abiotic, Atmosphere, Lithosphere,Hydrosphere, Biosphere		
1.4	Concept of sustainable development		

2.	Environmental Pollution	07 Hours	23%
2.1	Environmental pollution: types, causes, effects and controls; Air, water, soil and noise pollution		
2.2	Nuclear hazards and human health risks		
2.3	Solid waste management: Control measures of urban and industrial waste Pollution case studies – Ganga/Yamuna River,		
2.4	Bhopal Gas Tragedy, Delhi Air Pollution, Effect of Pandemics on the Environment		
3.	Ecology & Ecosystems	05 Hours	17%
3.1	Concept of an ecosystem, Structure and function of an ecosystem Producers, consumers and decomposers		
3.2	Energy flow in the ecosystem, Food chains, food webs and ecological pyramids		
3.3			
3.4	Case studies of the following ecosystems: Forest ecosystem, Grassland ecosystem Aquatic ecosystems (ponds, streams, lakes, rivers)		
4.	Biodiversity and Conservation	04 Hours	13%
4.1	Introduction – Definition: genetic, species and ecosystem diversity Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values		
4.2	Hot-spots of biodiversity, Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts.		
4.3	Endangered and endemic species of India, Conservation of biodiversity Biodiversity Act 2002/ BD Rule 2004: Mandate & Functions of National Biodiversity Authority (NBA),		
4.4	Role of State Biodiversity Board (SBB) and		
4.5	Biodiversity Management Committees (BMC).		
5.	Natural Resources	04 Hours	13%
5.1	Renewable and non-renewable resources		
5.2	Recyclable and Non-recyclable resources		
5.3	Energy resources: Growing energy needs, use of alternate energy sources. Case studies.		
5.4	Role of an individual in conservation of natural resources		
6.	Human Communities and the Environment	07 Hours	23%
6.1	Pandemics: Causes, Effects, Impact on the Environment: Positive & Negative, Lessons to learn		
6.2	Floods, Cyclones, Earthquakes, Landslides & Forest Fires		
6.3	Human population growth: Impacts on environment, human health and welfare.		
6.4	Case Studies: Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents, etc.		

Instructional method and Pedagogy:

The course is based on practical learning. Teaching will be facilitated by Slides Presentations, Reading Material, Discussions, Case Studies, Ted Talks, Videos, Task-Based Learning, Projects, Assignments and various Individual and Interpersonal activities like, Critical reading, Group work, Independent and Collaborative Research, Presentations, etc.

Evaluation:

There will be end semester university examinations based on Practical/ Viva. Students will be evaluated continuously in the form of internal as well as external evaluation. The evaluation is schemed as 25 marks for internal evaluation and 25 marks for external evaluation. The concerned teacher shall evaluate students and distribute the marks

Evaluation Scheme

The students' performance in the course will be evaluated through the following components:

Component		Marks
Internal	Attendance	05
	Assignment/ Individual Activity Participation/ Quiz, etc.	10
	Quiz	10
External	Group Activity Participation, Project Report, Case Study with Report, Online Certification Course, MOOCS, Field Work with report, Presentation etc.	10
	Viva/ Oral Examination	15
Total		50

Note: In internal component assignment and quiz are compulsory but in external component of 10 marks each subject teacher is free to conduct any activity mentioned above (that particular activity should be same for whole class).

Course Outcomes (COs):

At the end of the course, students will be able:

CO1	To perceive the elementary knowledge about natural environment and its relation with science.
CO2	To identify and analyze human impacts on the environment.
CO3	To understand the facts and concepts of natural and energy resources thereby applying them to lessen the environmental degradation.
CO4	To Initiate new and innovative environmental friendly practices.

CO5	To communicate on recent environmental problems thereby creating awareness among society.
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Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	3
CO2	-	-	-	-	-	-	3	-	-	-	-	-	-	3
CO3	-	-	-	-	-	2	3	-	-	-	-	-	-	3
CO4	-	-	2	-	-	-	3	-	-	-	-	-	2	-
CO5	-	-	-	-	-	-	-	-	-	2	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Text Books:

1. Varandani, N.S., Basics of Environmental Studies
2. Sharma, J. P., Basics of Environmental Studies

❖ Reference Books:

1. Shah Shefali & Goyal Rupali, Basics of Environmental Studies
2. Agrawal, K.C., Environmental Pollution : Causes, Effects & Control
3. Dameja, S. K., Environmental Engineering & Management
4. Rajagopalan, R., Environmental Studies, Oxford University Press
5. Wright Richard T. & Nebel Bernard J., Environmental Science
6. Shah, S.G., Shah, S.G. & Shah, G. N., Basics of Environmental Studies, Superior Publications, Vadodara

❖ Web Materials:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IITDelhi/Environmental%20Air%20Pollution/index.htm>
2. <http://nptel.iitm.ac.in/video.php?subjectId=105104099>
3. http://apollo.lsc.vsc.edu/classes/met130/notes/chapter1/vert_temp_all.html
4. <http://www.epa.gov>

5. <http://www.globalwarming.org.in>
6. <http://nopr.niscair.res.in>
7. <http://www.indiaenvironmentportal.org.in>
8. <http://nbaindia.org/>

PSUDI02: ENGINEERING PHYSICS - 2

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	50	-	50	

Outline of the Course:

Unit No.	Title of the Unit	Minimum number of Hours
Unit 1	Electricity and magnetism Lab	12
Unit 2	Light and Optics Lab	8
Unit 3	Modern Physics Lab	10

Total hours (Theory): 00

Total hours (Lab): 30

Total hours: 30

Detailed Syllabus:

1	Electricity and magnetism	12 hours (35%)
1.1	Charge, Conductors and Insulators, Coulomb's law, The electric field	
1.2	Principle of Superposition, Electric potential energy, Electric potential	
1.3	Capacitance and capacitors, The electric potential inside a parallel plate capacitor, electron current,	
1.4	Conductivity and resistivity, resistance and Ohm's law, Kirchhoff's laws and basic circuit, energy and power	
1.5	Magnetic field, The magnetic field of a current, magnetic dipoles	
1.6	Ampere's law and solenoids, Magnetic forces on current-carrying wires	
1.7	Magnetic properties of matter, Induced currents, Lenz's law, Faraday's law,	

1.8	Induced currents: Three Applications, Inductors, LC Circuits, LR Circuits, LRC Circuits	
No. of Experiment Measurement of capacitance by the bridge method Induction and LR, LC, and LRC Circuits Magnetic field along the axis of a coil Time Constant of RC Circuit Measurement of susceptibility of paramagnetic solution (Quinck's Tube Method)		
2	Wave and Optics:	08 hours (25%)
2.1	Classification of waves: Plane and Spherical Waves. Longitudinal and Transverse Waves. Plane Progressive (Travelling) Waves, Pressure of a Longitudinal Wave. Energy Transport	
2.2	Intensity of Wave, Standing (Stationary) Waves in a String: Fixed and Free Ends, Longitudinal Standing Waves and Normal Modes, Superposition of Waves, Propagation of electromagnetic waves	
2.3	Diffraction and Interference, reflection, refraction, refractive index,	
2.4	Basics of LASER Physics, Total Internal reflection, Basics of optical fiber, Acceptance angle and Numerical aperture	
No. of Experiment The wavelength of light, LASER, and Diffraction Numerical Aperture and Bending Losses in Optical fiber Standing (Transverse) Waves and resonance Using Vibrating Strings. Melde's Experiment		
3	Modern Physics:	10 hours (30%)
3.1	Planck's quantum theory, Planck's constant and light as a collection of photons	
3.2	Photo-electric effect and Compton scattering. De Broglie wavelength and matter waves	
3.3	Davisson-Germer experiment, Wave-particle duality, Heisenberg uncertainty principle- impossibility of a particle following a trajectory	
3.4	Two slit interference experiment with photons, atoms and particles; Radioactivity: stability of nucleus; Law of radioactive decay	
No. of Experiment The Photoelectric Effect; photo current versus intensity and wavelength of light; maximum energy of photo-electrons versus frequency of light Frank hertz experiment; To determine the ionization potential of mercury To determine value of Planck's constant using LEDs of at least 4 different colours		

Course Outcome (COs):

CO1	Students would be able to describe the static and dynamic electric and magnetic fields for technologically important structures.
CO2	Ability to identify and illustrate physical concepts and terminology used in optics and other wave phenomena.
CO3	Students would be able to appreciate the need for quantum mechanics, wave particle duality, uncertainty principle etc. and their applications.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	2	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	3	1	1	-	-	-	-	-	-	-	-	-	-
CO3	3	3	-	2	1	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
PSUD102	3.00	3.00	2.00	1.67	1.00	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, put “-”

Recommended Study Material:

❖ Text Books:

1. Physics for Scientists and Engineers by Randall D. Knight, 4th Edition, Pearson
2. University Physics by Hugh D. Young, Roger A. Freedman and A. Lewis Ford, 13th Edition, Pearson

3. Physics by John D. Cutnell & Kenneth W. Johnson, 8th Edition, John Wiley & Sons, Inc.

❖ **Reference Books:**

1. Electricity and Magnetism, Edward M. Purcell, 1986 McGraw-Hill Education
2. Waves: Berkeley Physics Course, vol. 3, Francis Crawford, 2007, Tata McGraw-Hill.
3. Fundamentals of Optics, F.A. Jenkins and H.E. White, 1981, McGraw-Hill
4. Principles of Optics, Max Born and Emil Wolf, 7th Edn. 1999, Pergamon Press.
5. Optics, Ajoy Ghatak, 2008, Tata McGraw Hill
6. Concepts of Modern Physics, Arthur Beiser, 2002, McGraw-Hill.
7. Introduction to Modern Physics, Rich Meyer, Kennard, Coop, 2002, Tata McGraw Hill
8. Introduction to Quantum Mechanics, David J. Griffith, 2005, Pearson Education.

❖ **Web Materials:**

1. Measurement of capacitance by the bridge method -
<https://www.youtube.com/watch?v=loZC-2A3LLg>
2. Phasor Diagram of RL, RC and RLC Circuits -
<https://www.youtube.com/watch?v=HaFrY0qQ-NU>
3. Magnetic field along the axis of a coil -
<https://www.youtube.com/watch?v=S0N4eVg7I3Y>
4. Quinck's Tube Method - <https://www.youtube.com/watch?v=yzgdq8uUfO4>
5. The wavelength of light, LASER - <https://www.youtube.com/watch?v=desLn3tMLcc>
6. Numerical Aperture and Bending Losses in Optical fiber -
<https://www.youtube.com/watch?v=b7dLcINlvwE>,
<https://www.youtube.com/watch?v=Wh9knsYSodI>

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

B.Tech (EC) Programme

A COURSE ON LIBERAL ARTS (SEMESTER-2)

HSUS101: PAINTING

Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
II	HSUS101	Painting	02	02	--	--	25	25	50

Course Objectives

- To encourage/ foster creativity among the students
- To introduce students to the fundamental processes of visual perception and artistic expression
- To cultivate / spawn awareness among students about the significance of arthistory, art criticism and aesthetics
- To help the students understand the meanings of concept, designs, shapes, colors, medium, and format
- To give the students the firsthand experience of design, painting, colors, light, shapes, shades and other important aspects of painting
- To develop in students an understanding of major styles and contemporary issues in painting

Course Outline

ModuleNo.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Painting An Introduction to Painting Principles of Composition Medium and Techniques of Painting History of Painting: Folk Indian Painting / Western Painting 2D and 3D Painting	2

2	Drawing from Nature and Object Objects of Drawing: Nature and Manmade / Artificial Objects Drawing Still / Live Objects Drawing from Memory Drawing from Life	4
3	Colour Design and Colour Value Color Theory:	6
	Color wheel (primary/secondary, complementary), transparency/opacity, hue, value (intensity, brightness), chroma (saturation, purity) & temperature (warm/cold) Color Contrast & Attributes: Interaction, harmony, psychology/mood, culture & expression Media Characteristics & Surfaces: Acrylic, oil, paper, wood & canvas (primed/unprimed)	
4	Composition and Perspective Composition: Space, movement, balance, asymmetry, rhythm, shapes, proportion & lighting Perspective: An approximate reproduction Types of Perspectives: Linear Perspective, One-point Perspective, Two-point Perspective, Three-point Perspective, Four-Point Perspective	6
5	Figure Drawing and Proportion Proportions of the Human Body Three views – Anterior (front), Lateral (side) and Posterior (back) Fundamental Proportion – The Big Three	4
6	Sketching Sketching and Freehand Sketching Techniques Sketch and Drawing Medium	4
7	Contemporary Issues in Painting Contemporary Indian Art Pioneers of Contemporary Indian Art Contemporary Issues in Painting	4
Total Hours		30

Instruction Method and Pedagogy

- Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation Scheme

- The evaluation scheme for the course will comprise the following components:
- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Major Learning Outcomes

- At the end of the course, a student will
- have cultivated a sense of creativity.
- be appreciative of art history, art criticism and aesthetics.
- be able to recognize the elements of arts in painting.
- have better cognizance and association of meaning of colors, shapes, and composition.
- be able to acknowledge the principles of painting as in design and colors, concept, medium and formats.
- have instantaneous painting experience about designing, lights, shades and colors and such other important aspects.

HSUS102: PHOTOGRAPHY

Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact	Theory		Practical		Total
				Hours/Week	Internal	External	Internal	External	
II	HSUS102	Photography	02	02	--	--	25	25	50

Course Objectives

- To introduce students to the tools and techniques of photography
- To provide students a thorough understanding of the mechanism and operations of a camera and help them understand the importance of aperture, shutter speed, film speed, depth of field, movement, and light meters to create a master shot
- To enable students to come out with a final project that demonstrates a single or a unified photographic idea or technique
- To explain students varied types of photographic representation including appropriation, persona, mixed media, non-objective images and engage them into experimentation using digital media
- To make students create a portfolio demonstrating creative uses of artificial and mixed lighting situations

Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Photography • Art, Design and Visualization • Basics of Photography and Various Types of Photography • Basics of Post Production • A Brief History of Photography: Early Experiments and Later Developments	03

2	Camera and Operating System • Role of Camera in the Photography • Types of Camera Pin-hole, box, folding, large and medium format cameras, single lens reflex (SLR) and twin lens reflex (TLR), miniature, subminiature and instant camera • Principal Parts of Photographic Camera Lens, Aperture, Shutters, various types and their functions, focal plane shutter and in-between the lens shutter, shutter synchronization, self-timer • Types of Lenses Single (meniscus), achromatic, symmetrical and unsymmetrical lenses, telephoto, zoom, macro, supplementary and fish-eye lenses • Different Models of Camera, their Features and Operating Systems • Camera and Size of the Image, Speed and Power of Lens	05
3	Light and Shade • Reflection and refraction of light • Dispersion of light through a glass prism, lenses • Colour Filters: Different kinds, Red, yellow, green, neutral density, half filters, filter factor, colour correction filter • Photographic Light Sources: Natural source, the Sun, nature and intensity of the sunlight at different times of the day, different weather conditions • Artificial light sources: Nature, intensity of different types of light sources used in photography namely; (i) Photo flood lamp, (ii) Spot light, (iii) Halogen lamp, Barn doors and snoot, lighting stands • Flash unit: Bulb flash and Electronic flash, main components, electronic flash units, studio flash, slave unit, multiple flash, computer flash, x-contact, exposure table	10
4	Composition • Different kinds of image formations • Principal focus and focal length of the lens • Depth of field, angle of view and perspective • Perspective and composition • Rules of composition	09
5	Contemporary Issues in Photography • Present Day Photography • Contemporary Photographers and their Contributions • Major Issues in Contemporary Photography	03
Total Hours		30

Instruction Method and Pedagogy

- Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation Scheme

- The evaluation scheme for the course will comprise the following components:
- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Learning Outcomes

- At the end of the course, students will be able to Understand, appreciate and demonstrate innovative approach, beauty and acute acumen in the area of photography
- Develop photography skills and become familiar with the functions and importance of the visual elements of nature and artificial objects
- Become independent thinkers who will contribute inventively and critically to culture through the making of art photography
- Have thorough understanding and acute sense of light and shade, composition, and presentation of a piece of art
- Experiment and Represent the cultivated sense and skills in Photography to the mass
- Prepare an impressive portfolio encompassing holistic approach to art and other the areas of study.

HSUSI03: SCULPTING

Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact	Theory		Practical		Total
				Hours/Week	Internal	External	Internal	External	
II	HSUSI03	Sculpting	02	02	--	--	25	25	50

Course Objectives

- To promote creativity and aesthetic sense pertaining to Sculpting by introducing them to the history of sculpting, its basic concepts and contemporary techniques and issues
- To help the students understand and develop the skill of sketching and drawing from life, natural and manmade objects and structures using various means like pencil, pen, ink, crayon, chalk, colour etc.
- To help them understand methods and materials of sculpture i.e. clay, plaster, cement, wood, stone, bronze, enlarging and reducing devices, welding torch etc.
- To help the students develop the sense of structure, and understand how forms achieve their structural unity through adherence to principles of physical nature of material being observed and studied (e.g. Plants, insects, minerals etc)
- To introduce the basic visual elements of 2-D and 3-D designs with emphasis on fundamentals of two and three-dimensional designs
- To acquaint the learners with various perspectives to draw and mould a sculpture
- To make the learners understand the colour theory and its practical usages
- To provide the students a sound background of the traditional and representational form in sculpture and enable them to develop their own vision

Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Sculpture What Sculpture is and how it is different from other Arts Basic elements, techniques, and history of sculpture Form, Space and Basic Shapes in Sculpture Casting materials, moldmaking, basic wood cutting and shaping, metal work, Welding, assembling, adhesives, plaster, mixing	06
2	Fundamental Principles of Sculpture Basic Principles of Aesthetics in Sculpture Visual Principles – balance, sequence, weight, and structural dynamics in sculpture Structural Principles and communicative possibilities of sculpture	10
3	Process of Modeling Additive and reductive processes Major Techniques of Sculpture: Modeling, Carving, Pointing Materials used in Modeling Clay Modeling and Carving	11
4	Contemporary Issues in Sculpture Sculpture and Present Day Context Contemporary Sculptors and their Contribution Major Contemporary Issues in Sculpture	03
Total Hours		30

Instruction Method and Pedagogy

- Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation Scheme

- The evaluation scheme for the course will comprise the following components:
- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

External Evaluation

- University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	✓	70	70
Total				70

Learning Outcomes

Upon successful completion of this course, students will be able to:

- Incorporate basic principles of aesthetics into sculpture

- Understand and apply basic concepts, styles and latest techniques of sculpting
- Explore traditional and experimental materials and design for sculpture
- Maintain a sketchbook of ideas and drawings to work out art project and to document coursework and discussions
- Understand the latest jargons, and develop collaborative skills to exhilarate the speed of accomplishing the piece of art
- Make their portfolio rich by accomplishing projects given during course
- Become familiar with varied key sculptural techniques and formal ideas through hands-on workshops and experimentation with a variety of materials and three-dimensional assignments
- Present their work with greater impact and confidence for future prospects
- Get benefitted into other subjects of their study by developing broader and all-inclusive approach to learning.

HSUSI04: POTTERY AND CERAMIC ART

Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact	Theory		Practical		Total
				Hours/Week	Internal	External	Internal	External	
II	HSUSI04	Pottery and Ceramic Art	02	02	--	--	25	25	50

Course Objectives

- To encourage/ foster creativity among the students
- To introduce students to the craft of clay working
- To make the students able to create as well as appreciate expressive, beautiful three dimensional clay forms
- To recognize and realize the physical, emotional, and spiritual benefits of working with clay, and to except and come to terms with clays humbling qualities
- To build a higher-level understanding of the ceramic process, creating an awareness of the benefits of clay as a useful tool in their art therapy studies and practices
- To develop a deeper knowledge of the ceramic process, and become more confident with their hand building and glazing techniques
- To help them communicate an idea or emotion through their artwork

Course Outline

ModuleNo.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Clay Elements and Materials of Ceramic Art and Shape Prehistoric Cultures Basics of Pinching, Slabbing and Coiling Potter's wheel, centring the clay, forming bowls and cylinders Trimming and burnishing on the wheel Sketching the Pottery Models	5
2	Types and Techniques of Making Pottery Types of Pottery: Porcelain Pottery, Earthenware Pottery, Stoneware Pottery Techniques of Pottery: Hand-Built Pottery: Pinch, Coil, Slab Wheel-Thrown Pottery	7
3	Methods of Making Pottery Coil Method Pinch Method Slab Method	8
4	Decorating the Clays Different Methods of Decoration Textures in Pottery Colours, Painting, Carving, Glazing etc. in Pottery	7
5	Contemporary Issues in Pottery and Ceramic Art Present Day Pottery and Ceramic Art Place of Pottery and Ceramic Art in Contemporary Art Society Major Practitioners of Contemporary Pottery and Ceramic Art and their Contributions	3
Total Hours		30

Instruction Method and Pedagogy

- Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

- The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical

evaluation. The evaluation (practical) is schemed as 25 marks for internal evaluation and 25 marks for external evaluation.

Evaluation Scheme

- The evaluation scheme for the course will comprise the following components:
- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Learning Outcomes

At the end of the course, the learners will

- Have basic understanding of clay and glaze composition and formulation with emphasis on hand built ceramic forms.
- Have explored a variety of hand building methods including extended pinch, slab built and extruded forms
- have learned firing and glazing methods for stoneware clay
- have learned how finishing and decorating contribute or detract from the intention as an artist
- Finally, a student will also have developed a sense of appreciation regarding how a unified, coherent form that is finely crafted is beautiful in its own right.

HSUS105: MEDIA AND GRAPHIC DESIGN

Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact	Theory		Practical		Total
				Hours/Week	Internal	External	Internal	External	
II	HSUS105	Media and Graphic Design	02	02	--	--	25	25	50

Course Objectives

- To encourage/ foster creativity among the students
- To introduce students to the fundamentals of graphic designs
- To cultivate / spawn awareness among students about the significance of art and designs, art criticism and aesthetics
- To help the students understand the meanings of concept, designs, shapes, colors, print and medium
- To give the students first-hand experience of working on Graphic Software
- To develop in students an understanding of major issues, techniques and aspects of designs and print

Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Media and Graphic Design Creating Art, Art in Context and Art as Inquiry History of Graphic Design Constructional, Representational, and Simplification Drawing	03

2	Layout and Design Layout, Design and Aesthetics Elements of Design Principles of Design: Harmony, Balance, Rhythm, Perspective, Emphasis, Orientation, Repetition and Proportion Impact/function of Design Indigenous design practices Role of design in the changing social scenario	07
3	Form and Space Types of Forms: Man-made, Nature Types of Space: Negative and Positive Composition of Form and Space to create Layout Exploring Creativity	06
4	Computer Graphics An Introduction to Graphic Software Flash, Coreldraw, Illustrator and Photoshop Pre-press Process	04
5	Fonts Construction of Type Anatomy of Type Visual Language Creating Logo and Symbol	04
6	Basic Print Media An Introduction to Press and its Development Phases Types of Press Types of Printing Technologies Post-press Processes	03
7	Contemporary Issues in Graphic Design Present Day Graphic Designs Contemporary Designers and their Contribution Major Contemporary Issues in Graphic Design	03
Total Hours		30

Instruction Method and Pedagogy

- Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

- The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 25 marks for internal evaluation and 25 marks for external evaluation.

Evaluation Scheme

- The evaluation scheme for the course will comprise the following components:
- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Learning Outcomes

At the end of the course, a student will

- have cultivated a sense of creativity.
- be appreciative of art and designs, art criticism and aesthetics.
- be able to recognize the elements of arts in graphic design.
- have better cognizance and association with the meaning of designs, shapes, colors, print and medium.
- be able to design graphics using computer softwares like Photoshop, CorelDraw, and Illustrator.

HSUS106: ART AND CRAFT

Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact	Theory		Practical		Total
				Hours/Week	Internal	External	Internal	External	
II	HSUS106	Art and Craft	02	02	--	--	25	25	50

Course Objectives

- To encourage / foster creativity among the students
- To enable the students to work through the process of bringing an idea from conception to realization
- To enable students to create artifacts that are visually expressive
- To cultivate / spawn awareness among students about the significance of art, craft and aesthetics
- To develop students' graphic skills which may work towards the realization of ideas in the creation of 2D and 3D
- To provide opportunities to students to be conversant with the use of a variety of materials, media, tools and equipments for Art and Craft

Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
I	An Introduction to Art and Craft Basic Design and Forms Space and Geometry Elements of Nature and Object	03

2	Paper Cutting Study of Designs Context of Space and Form Types of Textures and Papers Principles of Paper Cutting	05
3	Design from Nature Nature as a Source of Design Principles of Designing Natural Object Decorative Forms Cutting, Collaging, Embossing and Itching	08
4	Card Board Modeling Principles of Form and Space Dimensions of Space and Shape Process of Modeling and Decoration	06
5	Print Making An Introduction to Print Making A Brief History of Print Making Types of Print Making Processing of Print Making Sketching and Drawing	05
6	Contemporary Issues in Art and Craft Present-day Art and Craft Using the Waste for making the Best Contemporary Issues in Art and Craft	03
Total Hours		30

Instruction Method and Pedagogy

- Teaching will be practical - based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

- The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 25 marks for internal evaluation and 25 marks for external evaluation.

Evaluation Scheme

- The evaluation scheme for the course will comprise the following components:
- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Learning Outcomes

At the end of the course, students will be

- aware about the significance of art, craft and aesthetics.
- able to create artifacts that are visually expressive.

- able to lead the ideas from conceptualization with reference to the 2D and 3D model making.
- conversant with the use of a variety of materials, media, tools and equipments for creative Art and Craft.

HSUS109: DRAMATICS

Credits and Schemes

Sem.	CourseCode	Course Name	Credits	TeachingScheme	Evaluation Scheme				
				Contact Hours / Week	Theory		Practical		Total
					Internal	External	Internal	External	
II	HSUS109	Dramatics	02	02	--	--	25	25	50

Course Objectives:

- To acquaint students with the concept of performing arts
- To teach professional acting skills to the students
- To offer training in the key areas of performance
- To acquaint students with the history, theory, and aesthetic value of theatre
- To provide students extensive training in acting skills (monologues, dialogues, and group scenes from a variety of sources), dance, voice, theatre production, and rehearsal techniques, which culminates in a performance before a live audience

Couse Outline

Module	TITLE / TOPIC	Classroom Contact Sessions
1	Introduction to Drama Introduction to performing arts Drama - An art, a socializing activity, & a way of learning Form of Drama Elements of Drama Types of Drama	06
2	History of Drama and Contemporary Theatre Important world dramatists & drama—from Greek to modern Evolution of contemporary theatre in the context of developments in Indian theatre Major Movements in Drama	06
3	Theatre Design and Techniques Theatre Architecture Stage craft: Set, light, costume, make up, sound, props Theatre techniques: from selection of script to final performance	06

4	Technicalities of Stage Performance Selection of plot and character Improvisation Movement Voice, Speech, Imagination Character Development Scene Enactment	08
5	Contemporary Trends in Drama New Tendencies in theatre Drama and Society Using drama for Social Change and Education	04

Instruction Method and Pedagogy

- Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

- The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Evaluation Scheme

- The evaluation scheme for the course will comprise the following components:
- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Major Learning Outcomes

At the end of the course, a student will

- be aware about the concept of performing art and its nuances.
- display a working knowledge of historic of drama, its development and current trends in dramatics.
- demonstrate skills in the technical/design preparation and execution of a theatre performance.
- demonstrate the ability to work collaboratively.
- develop essential transferable skills in various relevant areas of the theatre.

HSUS110: CONTEMPORARY DANCE

Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact	Theory		Practical		Total
				Hours/Week	Internal	External	Internal	External	
II	HSUS110	Contemporary Dance	02	02	--	--	25	25	50

Course Objectives

- To introduce students to the concept of performing arts
- To develop in students the ability to express through the form of dance
- To foster creativity and innovativeness in students
- To enhance the aesthetic sensitivity among the students
- To inculcate in students contemporary dance techniques, philosophies, approaches, improvisation and performance disciplines
- To help students improve concentration, mental alertness, quick reflex action, physical agility and stress relief capacities
- To provide the students with complete awareness of one's own body
- To guide the students express themselves a natural way human feelings and expressions by creating harmony

Course Outline

ModuleNo.	Title / Topic	Classroom Contact Sessions
1.	Introduction to dance Dance as a Performing Art Dance as a Medium of Expression History and Development of Dance	4

2.	Types of Dance Western dance and classical dance Salsa, rumba, hip hop, tap dance, belly dance, etc. Indian Classical Dance forms: Odissi, Bharatanatyama, Kathak, Kathakali, Kuchipudi etc. Other Regional dance forms in India	6
3.	Basic Elements of Dance Movements of different parts of a body for Expression Concepts of: Nritya, Laya and Taal	4
4.	Technical Skills in Professional Contemporary Dance Dance technique: alignment, balance, co-ordination, flexibility and control Expressive / presentation skills: Dynamic energy, physical engagement with the given material and stage, etc. Skills and processes of rehearsal and production: physical energy, stamina and athleticism Musicality: clarity of timing and phrasing	6
5.	Contemporary Trends in Dance : Prevalent trends and techniques in contemporary dance Future trends in contemporary dance form On Stage Performance	10
Total Hours		30

Instruction Method and Pedagogy

- Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation Scheme

- The evaluation scheme for the course will comprise the following components:
- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Major Learning Outcomes

At the end of the course, a student will

- be able to develop ability to express through the form of dance.
- have enhanced aesthetic sensitivity.
- have improved concentration, mental alertness, quick reflex action, and physical agility.
- be able to express a natural way human feelings and expressions by creating harmony.
- be able to deliver contemporary dance performance.

HSUSIII: MUSIC (VOCAL)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	50	--	50	

Pre-requisite courses:

- No pre-requisites for this course

Objectives of the Course:

To facilitate the learners:

- To explore and understand Vocal Music
- with an opportunity to pursue the hobby of music on the campus
- to reduce stress and get energised to study better
- to explore and understanding about the Indian Music (Vocal), and its basic terms

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Vocal Music	05
2	Elements of Indian Vocal Music	08
3	Introduction to Ragas and Taals	08
4	A study of forms of Indian Vocal Music	05
5	Contemporary Trends in Indian Vocal Music	04

Total hours (Theory): --

Total hours (Practical): 30

Total hours: 30

Detailed Syllabus:

1.	Introduction to Vocal Music	05 Hours	18 %
	Definition of Music; Concept of Vocal Music; Basic history and evolution of Indian music; Understanding the basic concepts of Indian music, including Swara, Raga, and Tala		
2.	Elements of Indian Vocal Music	07 Hours	26%
	Sound (Dhwani) and its origin; Andolan and its types; Naad and its quality; Vibration and Frequency; Sahayak Naad, 10 Thats of Bhatkhande Ji, Saptak, Poorvang, Uttarang, Varna, Alankar, Vadi, Samvadi, Anuvadi, Vivadi, Gamak, Meend, knowledge of Taanpoora and its parts.		
3.	Introduction to Ragas and Taals	07 Hours	26%
	Learning basic ragas like Yaman and Bhupali; Understanding aroha (ascent) and avaroha (descent) of ragas; Introduction to the concept of tala (rhythm); Learning common talas like Teentaal and Ektaal; Basic tala exercises and rhythmic patterns.		
4.	A study of forms of Indian Vocal Music	05 Hours	16%
	Various forms of Indian Vocal Music: Carnatic Music, Thumari, Drupad etc; Study of Western Music and Style of Indian Music; Study of Regional Folk Music; Study of Gharana, Ravindra Sangeet and Carnatic Music		
5.	Contemporary Trends in Indian Vocal Music	04 Hours	14%
	Fusion Music; Indian Folk and Fusion; Sufi Revival; Cultural Fusion; Indian Vocal Music Festivals, Famous Indian Vocal Artists		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the concept and elements of Vocal Music
CO2	develop a clear and controlled singing voice with proper breath control and posture.

CO3	acquire the ability to sing basic scales, patterns, and simple compositions with accuracy and clarity.
CO4	understand and perform basic ragas and talas, including common rhythmic patterns.
CO5	appreciate and celebrate rich heritage of Indian Vocal Music

Evaluation Scheme

- The evaluation scheme for the course will comprise the following components:
- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Recommended study Material:

❖ Reference Books:

- "Ragas in Indian Music: A Complete Reference Source for Carnatic Ragas" by Dr. Rajan P. Parrikar - A comprehensive guide to Carnatic music ragas with historical context and audio samples.
- "Ragas of Indian Music: Their Structure and Evolution" by Nazir Ali Jairazbhoy - An exploration of the structure and evolution of ragas in Hindustani music.
- "The Raga Guide: A Survey of 74 Hindustani Ragas" by Joep Bor and others - A reference book with CDs that provides in-depth information on various Hindustani ragas, including their characteristics and performances.
- "Indian Music and the West" by Gerry Farrell - A study of the impact of Indian music on Western music and culture.
- "The Oxford Illustrated Companion to South Indian Classical Music" by Ludwig Pesch - A comprehensive resource on South Indian classical music, including Carnatic vocal music.

❖ Web Sources:

- https://onlinecourses.swayam2.ac.in/cec20_as04/preview
- <https://www.sharda.org/>
- https://onlinecourses.swayam2.ac.in/cec22_as02/preview
- <https://www.udemy.com/course/learn-indian-voice-music-from-the-scratch-to-advance/>

HSUSII2: MUSIC (INSTRUMENTAL) – TABLA

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/15	--	30	02
Marks	--	25/25	--	50	

Pre-requisite courses:

- No pre-requisites for this course

Objectives of the Course:

To facilitate the learners:

- with an opportunity to pursue the hobby of dance on the campus
- to reduce stress and get energised to study better
- to explore and understanding about the instrument – Tabla, and its basic terms
- to play basic tals on Tabla and learn improvisation

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
	Introduction to Music (Tabla)	04
	Basic Terms	06
	Main Varnas of Playing Tabla	08
	Basic Taals	08
	Improvisation in Tritaal	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to Music (Tabla)	04 Hours	13.33%
	Explanation of the various parts of Tabla Vilambit, Madhya and Dhrut Laya Different Bols/Syllables played on Tabla		

	Introduction of the procedure of playing Tabla		
2.	Basic Terms	06 Hours	20%
	Matra, Tal, Sam, Tali, Khali, Vibhag, Dugun, Avartan Sangeet, Nad, Swar, Laya, Bol, Kisme, Kayda, Mukhda, Thai, Tigun, Chaugun, Tukda		
3.	Main Varnas of Playing Tabla	08 Hours	26.66%
	Dhet, Gadi, Gana Kraan, KaRaam, Ghraan, GhaRaam, DhumakiTa, DhirDhir, TeTedheTe		
4.	Basic Taals	08 Hours	26.66%
	Trital, Jhaptal, Dadra Kaherva Rupak/Tewra Ektal/Adital		
5.	Improvisation in Tritaal	04 Hours	13.33%
	Tita and Tirkita kayda		

Evaluation Scheme:

The evaluation scheme for the course will comprise the following components:

Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks

Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

HSUSII3: MUSIC (INSTRUMENTAL) – GUITAR

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/15	--	30	02
Marks	--	50	--	50	

Pre-requisite:

- Correct posture for holding the guitar and plectrum

Objectives of the Course:

Students will be able to:

- Identify the different parts of a guitar song, such as the melody, harmony, and rhythm.
- Play a variety of strumming patterns and techniques.
- Understand the different types of scales and chords.
- Analyse arpeggios and riffs in different keys.
- Learn and perform contemporary guitar techniques and blues solos using a variety of scales and chords.
- Comprehend play guitar in a variety of styles, including acoustic, rock, blues, and country.

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to Structure, Tunes & Strumming	06
2.	Relation of Scales & Chords	06
3.	Arpeggios and Introduction to Riffs	07
4.	Contemporary Techniques and Blues Soloing	05
5.	Acoustic to Advance Style	06

Total hours (Theory): --

Total hours (Practical): 30

Total hours: 30

Detailed Syllabus:

1.	Introduction to Structure, Tunes & Strumming	06 hours	20%
	- Basic chord progressions and strumming, Counting the beats along with strumming, Playing along with the metronome. - Introduction to the notes on the guitar and tuning procedures, Different strumming patters and how you can create your own strumming patterns.		
2.	Relation of Scales & Chords	06 hours	20%
	- Sequencing patterns (Minor & Major), 3 notes per string patterns, practicing scales using backing tracks, Formation of minor scales, Minor scale relation with major scale and visualization of the shapes. - Harmonized major scale, Chord formation theory, Introduction to CAGED system., CAGED Barre chord shapes, Extended chord formation, Some basic shapes of extended chords.		
3.	Arpeggios and Introduction to Riffs	07 hours	22%
	- Arpeggios shapes using CAGED system, Arpeggio exercises with sweep picking. - Power chords, Power chord rhythm study using down strokes and accents, Palm Muting, Popular riffs.		
4.	Contemporary techniques and Blues Soloing	05 hours	18%
	- Slides, Hammer-on, Pull-off, Vibrato, Legato exercises, introduction to Bending - How 12 bar blues chord progressions are formed, Types of Blues chord progressions, Swing rhythm in 12 bar blues, Basic Blues riffs. - Pentatonic scale shapes, Pentatonic scale sequencing Patterns, How to use pentatonic scale in blues soloing.		
5.	Acoustic to Advance Style	06 hours	20%

	• Basic finger picking exercises, Etudes and popular song riffs, Fingerstyle rhythm. • Detailed study of modes and improvisation using modes, Extended arpeggios and application, more extended chords (used in Jazz) • Advanced blues soloing, Funk rhythm study, Advanced finger style grooves, Metal riffs and scale sequences.		
--	--	--	--

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	play a variety of tunes on the guitar, using different strumming patterns and techniques
CO2	understand and apply the relationship between scales and chords
CO3	play arpeggios and riffs on the guitar
CO4	learn and perform contemporary guitar techniques and blues solos
CO5	play guitar in a variety of styles, from acoustic to advanced
CO6	develop their own unique guitar sound and style

Recommended Study Material:

❖ Text book:

1. Alternate Picking – Handbook by Clifford Martin
2. Guitar: For Beginners by Nicolas Crater

❖ Web material:

1. <https://www.youtube.com/watch?v=wAfbTvEeMmw&list=PLovStLXqoeZIncXap2csJ-FKgLUCyPQzY>
2. <https://www.youtube.com/watch?v=rTMN8rCBWkw>
3. <https://www.coursera.org/learn/guitar>
4. <https://www.coursera.org/learn/guitar-scales-chord-progressions>

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

HSUSII4: MUSIC (INSTRUMENTAL)- HARMONIUM

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/15	--	30	02
Marks	--	50	--	50	

Pre-requisite courses:

- No pre-requisites for this course

Objectives of the Course:

To facilitate the learners:

- with an opportunity to pursue the hobby of dance on the campus
- to reduce stress and get energised to study better
- to explore and understanding about the instrument – Harmonium, and its basic terms
- to play basic Raags on Harmonium

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
6.	Introduction to Harmonium	04
7.	Basic Terms	06
8.	Swar	08
9.	Raag	08
10.	Tal	04

Total hours (Theory): --

Total hours (Practical): 30

Total hours: 30

Detailed Syllabus:

1.	Introduction to Harmonium	04 Hours	13.33%
	- Understanding the parts of the harmonium - How to set up and maintain the harmonium - Basic hand positioning and posture - Introduction to the keyboard layout and finger numbering		
2.	Basic Terms	06 Hours	20%
	- Swar, Raag, Tal, Alankar, Bandish, Saptak - Sthai, Aroh, Avroh, Aalap and Bolaan		
3.	Swar	08 Hours	26.66%
	- Major Scale (Sa, Re, Ga, Ma, Pa, Dha, Ni, Sa) - Shuddha, Komal and Teevra Swars - Basic Compositions of Seven Surs		
4.	Raag	08 Hours	26.66%
	- Kalyan - Bageshri - Bhairav - Bihag - Chhayanat - Puriya Dhanashri - Bhimpalas - Jaijaiwanti - Asawari - Bibhas - Miya Malhar - Multani - Darbari Kanada - Bilawal		
5.	Tal	04 Hours	13.33%
	- Trital - Ektaal - Chautaal - Dadra - Kaherva - Rupak - Tewra - Surfakta (Sul taal) - Deepachandi - Tilwada		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify the parts of Harmonium and basic positioning before playing
CO2	Use the basic terms with understanding when it is needed in Harmonium playing
CO3	Identify and play the basic Swars of Tabla playing
CO4	Play the basics of the Raags mentioned
CO5	Identify and play along with the Tals mentioned

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

HSUS115: MUSIC (INSTRUMENTAL) - FLUTE

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/15	--	30	02
Marks	--	50	--	50	

Objectives of the Course:

To facilitate the learners:

- To introduce students to the fundamental techniques and principles of playing the flute.
- To develop proficiency in basic flute playing skills, including tone production, finger technique, and breath control.
- To explore different musical styles and genres through flute repertoire.
- To foster an appreciation for the artistic and historical aspects of flute music.
- To encourage individual and ensemble performance experiences.

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to the Flute	04
2.	Fundamentals of Sound Production	06
3.	Finger Technique and Scales	06
4.	Flute Repertoire	06
5.	Flute Styles and History	04
6.	Ensemble Playing	04

Total hours (Theory): --

Total hours (Practical): 30

Total hours: 30

Detailed Syllabus:

1.	Introduction to the Flute	04 Hours	14%
	• Course orientation and objectives • Overview of the flute: parts and types • Assembly and disassembly of the flute • Proper posture and hand positioning • Introduction to basic embouchure		

2.	Fundamentals of Sound Production	06 Hours	20%
	- Breath control and support - Producing a clear and focused tone - Articulation techniques: tonguing and slurring - Dynamic control and expression - Tuning and intonation		
3.	Finger Technique and Scales	06 Hours	20%
	- Basic fingerings and hand coordination exercises - Major and minor scales - Arpeggios and scale patterns - Chromatic scale - Articulation exercises		
4.	Flute Repertoire	06 Hours	20%
	- Introduction to classical, folk, and popular flute repertoire - Selecting and preparing a solo piece - Interpretation and expression in music - Performance practice and stage presence		
5.	Flute Styles and History	04 Hours	14%
	- Historical overview of flute music - Major flute composers and their contributions - Exploring different styles: classical, jazz, world music, etc. - Listening and analysis of representative pieces		
6.	Ensemble Playing	04 Hours	12%
	- Introduction to flute ensembles - Collaborative rehearsal and performance - Role of the flute in chamber music - Developing ensemble communication skills		

Course Outcome (COs):

At the end of the course, the students will be

CO1	Demonstrate proper flute posture, hand position, and embouchure technique.
CO2	Produce a clear and controlled flute tone.
CO3	Execute basic fingering patterns and scales fluently.
CO4	Interpret and perform a variety of musical pieces on the flute.
CO5	Identify key historical and stylistic characteristics of flute music.
CO6	Participate in ensemble performances and collaborate effectively with other musicians.

Recommended Study Material:

❖ Reference books:

1. The Flute Book: A Complete Guide for Students and Performers by Nancy Toff

2. Trevor Wye's Practice Books for the Flute by Trevor Wye
3. Flute Basics: A Method for Individual and Group Learning by Sally Adams and Tracey Rush
4. The Art of Flute Playing by Edwin Putnik
5. Selected Flute Solos: Everybody's Favorite Series, Volume 101
6. The Flute and Flute-Playing by Theobald Boehm
7. Flute Music by French Composers edited by Louis Moyse

❖ Online Courses

1. Coursera - Introduction to Classical Music (Offered by Yale University)
2. Udemy - Flute Lessons for Beginners
3. Udemy - Intermediate Flute Lessons
4. edX - The Science of Flute Playing (Offered by MIT)
5. Skillshare - Flute: Start Playing Songs in 7 Lessons

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

HSUSII6: INDIAN CLASSICAL DANCE – KATHAK

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	50	--	50	

Pre-requisite courses:

- No pre-requisites for this course.

Objectives of the Course:

To facilitate the learners:

- with an opportunity to pursue the hobby of dance on the campus
- to develop a flair for performing arts.
- to acquaint students with Indian classical dance form, and its basic terms
- make them aware with basic mudras, abhinayas, and steps of the dance form.

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to Kathak	06
2.	Footwork and Tatkar	06
3.	Abhinaya (Emotive Expressions)	06
4.	Toda and Tukda (Compositions)	06
5.	Ghungroo (Ankle Bells) and Rhythmic Patterns	06

Total hours (Theory): --

Total hours (Practical): 30

Total hours: 30

Detailed Syllabus:

1.	Introduction to Kathak	06 Hours	20%
	- Understand the historical and cultural significance of Kathak. - Learn the basic postures (Tal Poses) and their importance. - Introduction to hand gestures (Mudras) and their meanings. - Develop a basic understanding of rhythm (Tala) in Kathak.		
2.	Footwork and Tatkar	06 Hours	20%
	- Learn and practice basic footwork patterns (Tatkar). - Develop proper foot placement and coordination. - Understand the importance of rhythm and synchronization in Kathak. - Explore the concept of Nritya (pure dance) in Kathak.		
3.	Abhinaya (Emotive Expressions)	06 Hours	20%
	- Understand the concept of Abhinaya in Kathak. - Learn and practice facial expressions (Bhava) and emotions. - Explore storytelling through dance.		
4	Toda and Tukda (Compositions)	06 Hours	20%
	- Learn and perform basic Kathak compositions (Toda and Tukda). - Understand the structure of Kathak compositions. - Practice and refine the choreography.		
5.	Ghungroo (Ankle Bells) and Rhythmic Patterns	06 Hours	20%
	- Explore the significance of Ghungroo (ankle bells) in Kathak. - Learn to wear and produce rhythmic sounds with Ghungroo. - Practice and execute various rhythmic patterns (Bols). - Understand the importance of laya (tempo) in Kathak.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	remember basic postures and hand gestures of Kathak.
CO2	understand the concept of Nritya in Kathak.
CO3	design storytelling through dance and will grasp the concept of Abhinaya and its role in Kathak.
CO4	perform basic Kathak Toda and Tukda compositions.
CO5	apply and create rhythmic sounds effectively with Ghungroo executing various rhythmic patterns through laya in Kathak.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

HSUSII7: INDIAN CLASSICAL DANCE – BHARATANATYAM

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	50	--	50	

Pre-requisite courses:

- No pre-requisites for this course.

Objectives of the Course:

To facilitate the learners:

- with an opportunity to pursue the hobby of dance on the campus
- to develop a flair for performing arts.
- to acquaint students with Indian classical dance form, and its basic terms
- make them aware basic mudras, abhinayas, and steps of the dance form.

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to Bharatanatyam	06
2.	Adavus and Footwork	06
3.	Abhinaya (Emotive Expressions)	06
4.	Rasa and Bhava	06
5.	Choreography and Performance	06

Total hours (Theory): --

Total hours (Practical): 30

Total hours: 30

Detailed Syllabus:

1.	Introduction to Bharatanatyam	06 Hours	20%
	- Understand the historical and cultural significance of Bharatanatyam. - Learn the basic postures (Asanas) and their importance. - Introduction to hand gestures (Mudras) and their meanings. - Develop a basic understanding of rhythm (Tala) in Bharatanatyam.		
2.	Adavus and Footwork	06 Hours	20%
	- Learn and practice basic Adavus (fundamental dance steps). - Develop proper footwork and coordination. - Understand the importance of rhythm in Bharatanatyam. - Explore the concept of Nritta (pure dance) in Bharatanatyam.		
3.	Abhinaya (Emotive Expressions)	06 Hours	20%
	- Understand the concept of Abhinaya in Bharatanatyam. - Learn and practice facial expressions (Bhava) and emotions. - Explore storytelling through dance.		
4	Rasa and Bhava	06 Hours	20%
	- Explore the nine Rasas (emotions) in Bharatanatyam. - Learn how to evoke different emotions through expressions and movements. - Understand the importance of emotional connectivity in dance.		
5.	Choreography and Performance	06 Hours	20%
	- Learn a simple Bharatanatyam composition. - Understand the structure of a dance performance. - Practice and refine the choreography. - Gain confidence in performing in front of an audience.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	understand the historical and cultural significance of Bharatanatyam.
CO2	analyse proper footwork and coordination.
CO3	design storytelling through dance.
CO4	perform to evoke different emotions through expressions and movements.
CO5	remember the choreography and gain confidence in performing in front of an audience.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

B.Tech (EC) Programme

VALUE ADDED COURSES (SEMESTER 1 & 2)

***CUUVI01: COMMUNITY ENGAGEMENT AND SUSTAINABLE
DEVOLVEMENT (CESD)***

Credits	Course Duration (Hours)	Course Span
2	45	One semester (1 st /3 rd)

Mode	Week	Hours	sessions
Interactive Mode	15	30	15
Practical mode	1(2 Days)	10	-
Reflective Mode	5	5	-
Total	21	45	15

Session Planning for CESD in Three Modes

1. Interactive Mode

- Duration: 30 hours
- Activities:
 - Interactive dialogues
 - Group discussions
 - Role-playing scenarios
 - Expert lectures
- Objectives:
 - Enhance understanding of SDGs and community engagement.
 - Develop communication and management skills.

2. Practical Mode

- Duration: 10 hours
- Activities:
 - Fieldwork and community visits
 - Civic, social, nature, health, and education awareness programs
 - Research and surveys

- Objectives:
 - Apply theoretical knowledge to real-world scenarios.
 - Engage directly with communities and understand their needs.

3. Reflective Mode

- Duration: 05 hours
- Activities:
 - Case studies analysis
 - Report writing and submissions
 - Reflection sessions
- Objectives:
 - Analyze and document experiences and learnings.
 - Reflect on the impact of community engagement activities.

This structured approach ensures comprehensive learning, practical experience, and reflective thinking, equipping students with the necessary skills for community engagement and sustainable development.

1. Interactive Mode

Students will participate in a variety of activities such as personal hygiene sessions, cleanliness drives, involvement in the Fit India Movement, awareness campaigns, creative tasks like poster making and slogan writing, essay competitions, discussions under the theme Let's Talk Green, organizing awareness rallies and others.

Sr.No	Week	List of Suggestive Activities	Duration
1	1 Week	Webinar on Personal Hygiene	2 Hours
2	2 Week	Seminar for fit India Movement	2 Hours
3	3 Week	Youth survey for National Youth Policy	2 Hours
4	4 Week	Expert Session on voters Awareness	2 Hours
5	5 Week	Seminar on water resources management	2 Hours
6	6 Week	Maths Science Model Training	2 Hours
7	7 Week	Poster Making , Slogan Writing, Essay Writing sessions	2 Hours
8	8 Week	Art & Craft Training to children of rural Area	2 Hours
9	9 Week	Webinar on 'Let's Talk Green'	2 Hours

10	10 Week	Webinar on Effective use of PPE Kit	2 Hours
11	11 Week	Cleanness Drive on Campus	2 Hours
12	12 Week	Expert lecture on Civic Awareness	2 Hours
13	13 Week	Awareness Rally on Addiction	2 Hours
14	14 Week	Street Play on Social Issues	2 Hours
15	15 Week	Tree Plantation	2 Hours

- Above mentioned are suggested topics , respective PO- CC may add upon as per their need and execute accordingly

2. Practical Mode

Sr.No	Two Days Workshop	Hours
1	Cleanness Drive in Villages, Education Amelioration Activities in Rural Schools, Awareness Drive on diseases	5 Hours
2	NO Plastic Use Awareness Drive –Rally etc. Education Awareness by door to door Campaign, Survey for government schemes	5 Hours

3. Reflective Mode

Sr.NO	Self-Development Skill	Hours
1	Assignments & Reports For All Above Activities , Self-guided Activities	05 Hours

Institute-Wise Semester Schedule for Offering CESD

Sr. No	Institute	Semester offered	Mode	Remarks
1	IIIM,CMPICA,PDPIAS,ARIP,MTIN,BDPIS	First	Three mode	
2	CSPIT and DEPSTAR	Second	Workshop mode	
3	RPCP	Third	To be decided	

Department / Institute wise Subject Coordinator Details 2024-25				
Sr. No.	Name of Institute	Programme officer Name	Mobile Number	Email ID
1	PDPIAS	Dr. Rajesh Savalia	9429222899	rajeshsavalia.maths@charusat.ac.in
2	ARIP	Shreya Swami	6353651313	shreyaswami.phy@charusat.ac.in
3	MTIN	Hetal Shah	9033404171	hetalshah.nur@charusat.ac.in
4	CMPICA	Hardik Pandit	9426561926	hardikpandit.mca@charusat.ac.in
5	RPCP	Hardik Koria	9731835384	hardikkoria.ph@charusat.ac.in
6	BDPIS	Parth Thakor	9998978550	parththakor.cips@charusat.ac.in
7	IIIM	Dr.Poonam Amrutiya	9429558484	poonamamrutia.mba@charusat.ac.in
8	DEPSTAR-IT	Hitesh Makwana	8460752501	hiteshmakwana.dit@charusat.ac.in
9	DEPSTAR-CE	Kashyap Patel	9974196044	kashyappatel.dce@charusat.ac.in
10	DEPSTAR-CSE	Gaurang Patel	9537433667	gaurangpatel.dcs@charusat.ac.in
11	CSPIT-CIVIL	Kanchal Dave	9978450993	kanchaldave.cv@charusat.ac.in
12	CSPIT-ME	Satayu Travadi	9723833357	satayutravadi.me@charusat.ac.in
13	CSPIT-EE	Ankur Patel	9978782503	ankurpatel.ee@charusat.ac.in
14	CSPIT-EC	Dr. Sagar Patel	7874635764	sagarpatel.ec@charusat.ac.in
15	CSPIT-CE	Martin Parmar	8488065457	martinparmar.ce@charusat.ac.in
16	CSPIT-IT	Rajnik Katriya	9913617724	rajnikkatriya.it@charusat.ac.in
17	CSPIT-CSE + AIML	Pinal Hansora	9737174610	pinalhansora.cse@charusat.ac.in

CUUVI02: PHYSICAL EDUCATION AND SPORTS

Credits	Course Duration (Hours)	Course Span
2	45	One semester

Mode	Week	Hours	sessions
Regular mode	15	30	15
Workshop mode	2 Days	10	-
Self-learning	1 day	05	-
Total	15 (3days)	45	15

TEACHING LEARNING STRATEGIES

The content will be taught by using lecture, demonstration, explanation, presentation methods, videos, learning by doing, Whole part whole method, Drills

COURSE LEARNING OUTCOME

After completing this course, the students will be able to

- Acquire, analyse and interpret basic skills
- Appraise the rules and regulation.
- Exhibit and assess a range of fundamental abilities, method and gaming strategies.
- Improve their physical fitness, including aspects such as strength, speed, agility, and endurance, to enhance their performance on the field.
- Enhance their ability to make effective decisions under pressure, including when to during the game.
- Understand the importance of sportsmanship, respect for opponents, referees, and teammates, and adherence to fair play principles.
- Learn mental preparation techniques to enhance focus, concentration, confidence, and resilience in competitive situations.
- Learn techniques for injury prevention, including proper warm-up, cool-down, and recovery strategies.

One Semester Course (Regular mode)

PLA N NO	SESSION TOPIC Football	SESSION TOPIC Basketball	SESSION TOPIC Volleyball	SESSION TOPIC Kabaddi
1	Football fundamental/Passing	Fundamental Basketball /Passing & Receiving	Volleyball fundamental/passing	footwork and agility training
2	Receiving/Controlling	Dribbling & Ball Handling	Service training	Simple raid
3	Dribbling	Controlling & Dribbling	Setting	Bonus point
4	Shooting	Passing & Catching	Blocking	Running hand-touch
5	First touch	Shooting / Jump Shot	Attack training	Toe touch
6	Passing and Supporting	Lay ups Shot	Defensive/offensive skill training	Kick training
7	Long ball /Air ball	Rebounding & Stealing	Footwork and agility training	Cover block
8	Linkup and Coordination	Defending (Defense)	Advance digging training	Single block
9	Set piece (penalty kick, corner kick) (free kick) (Throw in)	Attacking (Offense)	Position-specific drill training	Ankle hold
10	Dribbling (1vs1) (1vs2)	Off-ball Movements	“w” formation / 5-1 Defence system	Back hold
11	One touch passing	Two-on-Two and Three-on-Two	Team drills –rotational drills	Knee hold
12	Defending	Blocking & Intercepting	Combination drills session	Double thigh-hold
13	Attacking	Inbounding the Ball	Tactical session	Team drills-communications, rotational
14	Weight of the ground pass	Fast Break	Peak performance	Kabaddi communication drills
15	Shadow game(match style)	Tactical & Technical Skills	Tournament preparation	Game like drills

Workshop mode- Two Days

Workshop Duration	Football	Basketball	Volleyball	Kabaddi
Two Days	Official match	Official match	Official match	Official match
	Official match	Official match	Official match	Official match

Self-learning mode

Course Period	SELF -LEARNING MODE	Particulars
1 semester	Football	Case Studies or Project of Matches/ Players / World Tournaments /Book/Survey
	Basketball	
	Volleyball	
	Kabaddi	

A self-learning this course aims to develop a range of knowledge areas to improve students' understanding and performance in the sport. Here are the potential outcomes:

- Self-Paced Learning
- Resource Utilization
- Better Psychological Skills
- Enhanced Teamwork and Communication
- Knowledge of Rules and Regulations

This course execution Sports Faculty:-

Games	Football	Basketball	Volleyball	Kabaddi	Yoga
Faculty	Harsh Parmar	Ajay Raval	Pinkal Gamit	Tejal Gamit	External expert
Venue	Football Ground	Basketball court	Volleyball Ground ,New Sports Complex	Kabaddi Ground , New Sports Complex	

FTXXXX: SKILL AUGMENTATION

Preamble:

Industry needs professionals who can work successfully in teams, who have leadership qualities, who are alive to social and community needs and who can bring innovation and creativity to their work. Hence, to become successful professionals, students should have excellence soft skills, leadership qualities, team spirit, entrepreneurial capabilities and societal commitment besides expertise in their chosen fields.

Therefore, in order to prepare students to match these multifarious requirements, it is proposed to offer a 2 credit course on “Skill Augmentation” and implement evaluation system of the proposed course based on '100 Activity Points' for the award of engineering degrees in Faculty of Technology and Engineering, CHARUSAT. Under this system, engineering students will have to secure at least **100 minimum Activity Points** during the program duration to earn a B. Tech. degree over and above the academic grades. Each activity carries a specific number of points based on its significance and level of involvement. The students will be granted points on the basis of their participation in extra-curricular and co-curricular activities such as technical quizzes, tech-fest, paper presentation, rural internship, poster presentation and cultural activities, etc.

It is envisaged that the students will be able to reap benefits from these activities at their own pace and comfort. It is expected that by the time, students reach their Final Year, they would have developed themselves so well both through their studies in the respective technological field and through their active participation in the co-curricular and extra-curricular activities that they would be well-prepared for contributing to building the India and the world of their dreams.

It was emphasized that this course could help in fuller realization of the Graduate Attributes laid down by the university as well as could serve as an important ingredient facilitating the progression and recognition of the fast learners.

Mandatory Earning of Activity Points:

An Undergraduate Engineering student must earn the following minimum Activity Points as per the level of Entry in the B.Tech. Program as shown in **Table – 1**.

Table 1: Minimum Activity Points as per the level of Entry in the B.Tech. Program

Sr. No.	Students Category	Total Years for Points	Minimum Activity Points prescribed by AICTE
1	Regular students admitted to the 4 years Degree program	1 st to 4 th Year	100
2	Students entering 4 years Degree program through lateral entry	2 nd to 4 th Year	75
3	Students transferred from other Universities to the fifth semester	3 rd to 4 th Year	50

These points must be earned based on active participation in co-curricular and extra-curricular activities spanning through all the semesters of study of UG Program. Every student may choose, as per his/her liking, activities to achieve the mandatory points before becoming eligible for award of the Degree. The activities can be spread over the years of the duration of the program, anytime during the semester weekends and holidays, as per the interest and convenience of the student from the year of entry to the program. Activities must be selected from prescribed activity point table given in **annexure I**.

Minimum Earning of Activity Points

Regular students admitted to the 4 years Degree program is required to accumulate a minimum of 25 points annually in order to fulfil the criteria of attaining 100 points for the conferment of a B. Tech Degree.

Students entering 4 years Degree program through lateral entry (D2D Student) is required to accumulate a minimum of 25 points annually from 2nd year onwards to fulfil the criteria of attaining 75 points for the conferment of a B. Tech Degree.

Every student has to do minimum one activity from each category mention in **Activity Point Table** as per **Annexure 1**.

Students opted for early exit will be given Skill Augmentation certificate indicating number of activity points earned by the student.

Guidelines for the students:

Every student shall participate in the co-curricular and extra-curricular activities as prescribed in **Activity Point table** as per **annexure 1**.

Students will be provided a certificate duly signed by the endorsing authority of University as mention in **annexure 1**.

In every semester, every student is required to prepare an **Activity Point file** containing documentary proofs of activities, done by him/ her. Each student shall keep with him/her the **Activity Point file** duly signed by his/her counsellor and HOD.

Student is required to upload each activity with proof in CHARUSAT E – Governance system.

Students should earn the required points before appearing for 8th Semester examinations.

A student's result of his/ her Final Semester Examinations will be withheld until he/she completes the minimum Activity Points by the end of his/her B.Tech. Program.

In every semester, coordinator (s) shall conduct academic auditing of co-curricular and extra-curricular activities available for students, the monitoring mechanism of Activity Points to be earned by the students.

Evaluation Guidelines:

Course Coordinator will award the Activity Points for each student at the end of every Semester, after conducting the Viva Examination.

There will be neither grades nor marks for these points.

Absenteeism at the Viva examination for Activity Points at the end-semester examination will lead to zero accretion to the Activity Points, earned by the student.

All Activity Points have only Viva Component. Viva will be conducted to validate student claims. It is compulsory for every student to prepare five minutes of presentation for activities carried out during current semester.

The Exam fees is according to CHARUSAT norms for viva exam.

Two credits are given for this on a pass/ fail basis and is mandatory for getting the B. Tech Degree. As the Activity Points and credits are not based on marks, this won't be included in the Cumulative Grade Points Average (CGPA) but mandatory for the award of the Degree.

The credit earned shall be reflected on the student's eighth semester grade card.

In case students fail to earn the prescribed Activity Points, an eighth semester grade card shall be issued only after earning the required Activity Points.

Students shall be admitted for the award of degree only after the release of the eighth semester grade card.

Any concerns raised by the students regarding Activity Points shall be looked into in the combined meeting of HoD, Department Coordinator, Course Coordinator and the students concerned. The Principal/ HoD shall ensure the proper redressal of the concerns raised by the students regarding Activity Points. The Course Coordinator shall keep a hard copy of the consolidated statement of Activity Points of the students.

Teaching and Examination Scheme

Table 2: Teaching and Examination Schemes

Semester	Course Code	Course Title	Teaching Scheme		Examination Scheme			
			Contact Hours	Credit	Theory		Practical	
					Internal	External	Internal	External
8	FTXXX	Skill Augmentation	200	2	--	--	100 Activity Points	--

Best Achievers

The best 3 students will be selected from each department, who are doing thorough efforts, will be awarded/appreciated in a special function.

ANNEXURE-I: ACTIVITY POINT FRAMEWORK

Major category	Sub-category	Nature of Participation	Activity level and Points				Proposed Endorsing authority/ body @CHARUSAT	Valid Documentary Evidence
			U	S/ Z	N	IN		
Category -1 Outstanding Contributions in Research/ Innovation/ Entrepreneurship/IPR Graduate Attributes aligned: 1) Academically Excellent 2) nterdisciplinary knowledge	SSIP/Start-up/ Innovation ventures	Incubation at Proof-of-Concept Stage	30				CHARUSAT Start-up and Innovation Center (CSIC)	Certificate/ Recognition Letter/Relevant Legal Document
		Company Registered	40					
		Extramural fund received/ Revenue Generated	50					
	Patents/Copyrights	Filed	20				IPR Cell	
		Published	50					
		Granted	75					
	Product/Services development	Prototype developed	50				CIVF	
	Research Publications in WoS/Scopus listed journals)	Main Author	-	-	30	40	Corresponding Author(Faculty) / URC	Manuscript and endorsement from corresponding
		Co-Author	-	-	15	20		
	Participation in Technical/ Scientific Events	Poster / Oral Paper Presentation in Scopus / WOS Index Conference online/offline	10	12	16	20	HoD/Principal; endorsed by Faculty coordinator/Conve ner of Event if University level	Certificate
		Project Presentation /Innovation idea presentation / Model etc.)	10	12	16	20		
		Tech event participation (seminar, workshop, webinar, technical session etc.)	4	6	8	10		

Category 2 Academic Skill & Ability Enhancement Initiatives Graduate Attributes aligned: 1) Academically Excellent 2) Individually Excellent and Committed	MOOCs (other than curriculum) (Swayam/NPTEL)	04 -08 weeks	20				HoD/Principal	Final assessment certificate
		More than 08 weeks	30					
	MOOCs (other than curriculum) on Online Learning Platform like (Coursera)	04 -08 weeks /Min. 30 Hrs	5				HoD/Principal	Final assessment certificate
		More than 08 weeks / Min 50 hours	10					
	Industry Recognized Certification Courses	-	30				HoD/Principal	Final assessment certificate
	Summer/Winter Schools (other than curriculum)	Minimum 1 Week	10	20	30	40	HoD/Principal	Certificate/Endor sement/Selection
Competitive Examinations (GATE/CAT/GMAT etc)	Qualification	-	-	30	40	HoD/Principal	Recognition letter/Result document	
Category 3 Leadership, Institutional Management and Value-additions: Graduate Attributes aligned: 1) Attuned to cultural diversity 2) Active Global Citizens 3) Individually Excellent and Committed 4) Leaders in	Institute/ University Organized Event Management	Coordinator	16	20	25	30	Central Council / HOD/ Principal	Nomination letter/ Authorization letter; Copy of Magazine
		Co-coordinator(s)	12	16	20	25		
		Volunteers/Members	8	12	16	20		
	Department Organized Event Management	Coordinator	12	20	25	30	Central Council / HOD/ Principal	Nomination letter/ Authorization letter; Copy of Magazine /
		Co-coordinator(s)	8	15	20	25		
		Volunteers/Members	5	10	15	20		
	Representation in cultural/sports activities	Player/Performer	10	20	30	40	Central Council / HOD/ Principal	Certificate

General Note:	Point augmentation scheme for events					
Awards/Recognition/Outstanding contribution: points to augmented to the above	Winner-3rd position (points to be augmented)	4	8	12	16	U - University S/Z - State/Zonal N - National IN - International
	Winner-2nd position (points to be augmented)	6	10	14	18	
	Winner-1st position (points to be augmented)	8	12	16	20	
	top 5% in NPTEL	-	-	12	-	
	top 2% in NPTEL	-	-	14	-	
	top 1% in NPTEL	-	-	16	-	

1. Category-wise Participation Requirement:

Every student must complete at least one activity from each of the three categories as listed in the Activity Points framework.

2. Weightage for Categories 1 & 2:

A minimum of 50% of the total required points (i.e., 50 out of 100 points) should be earned through activities falling under Category 1 (Research/Innovation/IPR) and Category 2 (Academic Skill & Ability Enhancement).

EXIT COURSE

ECUS110: PCB DESIGNING AND TESTING

2nd Semester and 1st Year
(For Exit Option after First Year)
B. Tech. (EC)

Prerequisites: Basic knowledge of Physics

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Foundation of PCB designing concepts	8
2.	Component their categories understanding	16
3.	Introduction to PCB development tools	24
4.	Detailed description and practical of PCB designing	24
5.	Designing concepts and hands on practice	24
6.	Assembly Process View	12
7.	Wave Soldering Overview and Factors Affecting Wave Soldering	12
	Total hours (Lab):	120
	Total hours:	120

Detailed Syllabus:

1	Foundation of PCB designing concepts	08 Hrs	8%
What is PCB? Layers in PCB, Single Sided PCB, Double Sided PCB, Multi-Layer PCB, Trends in PCB Design, and Etching Process. ● To observe the etching process			
2	Component their categories understanding	16 Hrs	12%
Types of components, various types of packages of components, SIP (Single Inline Package), DIP (Dual Inline Package), Through Hole Packages, Surface mount packages, Soldering and soldering of components. ● Hands on sessions for Soldering and soldering of components			
3	Introduction to PCB development tools	24 Hrs	20%
Introduction of PROTUEUS ISIS and ARES software for schematic entry and PCB designing, Selecting the Components Footprints as per design, Picking and placing the Component, Making New Footprints, Assigning Footprint to components ● Simulate electronics components in simulation software.			
4	Detailed description and practical of PCB designing	24 Hrs	20%
Net listing, PCB Layout Designing, Prototype Designing, Design Rule Check (DRC), Drilling, Assembly of components ● PCB layout design in simulation tool			

5	Designing concepts and hands on practice	24 Hrs	20%
Introduction to Auto routing, Footprint, Pad stacks, Tracks, Track Size, Making the schematic for Academic and Industrial projects, PCB Design, Physical creation of PCB ● Create footprint in PBD designing tool			
6	Assembly Process View	12 Hrs	10%
Overview of the PCB assembly process, components placement, through-hole vs. surface mount technology.			
7	Wave Soldering Overview and Factors Affecting Wave Soldering	12 Hrs	10%
Introduction to wave soldering, key factors affecting the process such as solder wave height, conveyor speed, preheat temperature, and flux application.			

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	To learn and explore different aspect of Printed Circuit Board Design and fabrication.
CO2	To understand various kinds of PCBs, Entry Rules for Schematic, Component Layout methods, and Schematic Design, Placement Rules, Routing Techniques for Single Sided Board, Post Processing of design and Fabrication documents.
CO3	To design and fabricate PCBs.
CO4	Understand baseline process parameters

Course Articulation Matrix:

	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CO1	3	2	1	1	2	1	-	-	2	2	1	1	3	2
CO 2	3	3	2	2	3	2	1	1	2	2	2	1	3	2
CO 3	3	3	3	2	3	2	1	1	2	2	2	1	3	2
CO 4	3	3	2	3	3	2	-	-	2	2	2	1	3	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Reference Books:

1. "Printed Circuit Board Designer's Reference" by Christopher T. Robertson, ISBN: 9780130674814, Prentice Hall Professional Technical Reference, 2004.
2. "PROTEUS PCB Design Examples - ISIS & ARES", by George Shopov, Kindle Edition, 2020.
3. Coombs, Clyde F., Jr., and Happy T. Holden, eds. 2016. Printed Circuits Handbook. 7th ed. New York: McGraw-Hill Education.
4. The Handbook of Machine Soldering: SMT and TH 3rd Edition, Kindle Edition by Ralph W. Woodgate

❖ Web material:

- https://www.pcbway.com/blog/PCB_Design_Tutorial/
- <https://learn.sparkfun.com/tutorials/pcb-basics/all>
- <https://www.allaboutcircuits.com/technical-articles/pcb-design-tutorial/>
- <http://www.pcb-solutions.com/>

❖ Software:

- PROTEUS
- EasyEDA
- KiCAD